

Secure email server setup using Postfix, Dovecot, TLS, spam filtering, and authentication

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Abstract:

This project presents the design, deployment, and validation of a secure email server infrastructure based on Postfix and Dovecot on a Linux virtual private server.

The system was implemented using Postfix as the Mail Transfer Agent (MTA) and Dovecot as the IMAP server, with virtual mailboxes for domain users. Transport Layer Security (TLS) was enabled using Let's Encrypt certificates to protect SMTP and IMAP communications. Email authentication mechanisms including SPF, DKIM, and DMARC were configured to ensure message authenticity and improve deliverability. SpamAssassin was integrated into the mail flow to detect and filter unsolicited messages using rule-based and Bayesian analysis. User access and mail flow were tested using the Thunderbird email client.

The deployed server successfully handled authenticated mail submission, encrypted mail retrieval, internal and external email delivery, and spam classification. Verification against external providers (Gmail) confirmed correct SPF, DKIM, and DMARC alignment. The results demonstrate that a fully functional and secure email system can be implemented using open-source components when configuration, DNS alignment, and authentication are correctly applied.

Stack:

- Postfix — Mail Transfer Agent (send/receive mail)
- Dovecot — IMAP server
- SSL/TLS (Let's Encrypt)
- SpamAssassin — Filter spam before it reaches user mailboxes
- SASL — Authenticate users securely for sending/receiving mail

- Thunderbird — GUI

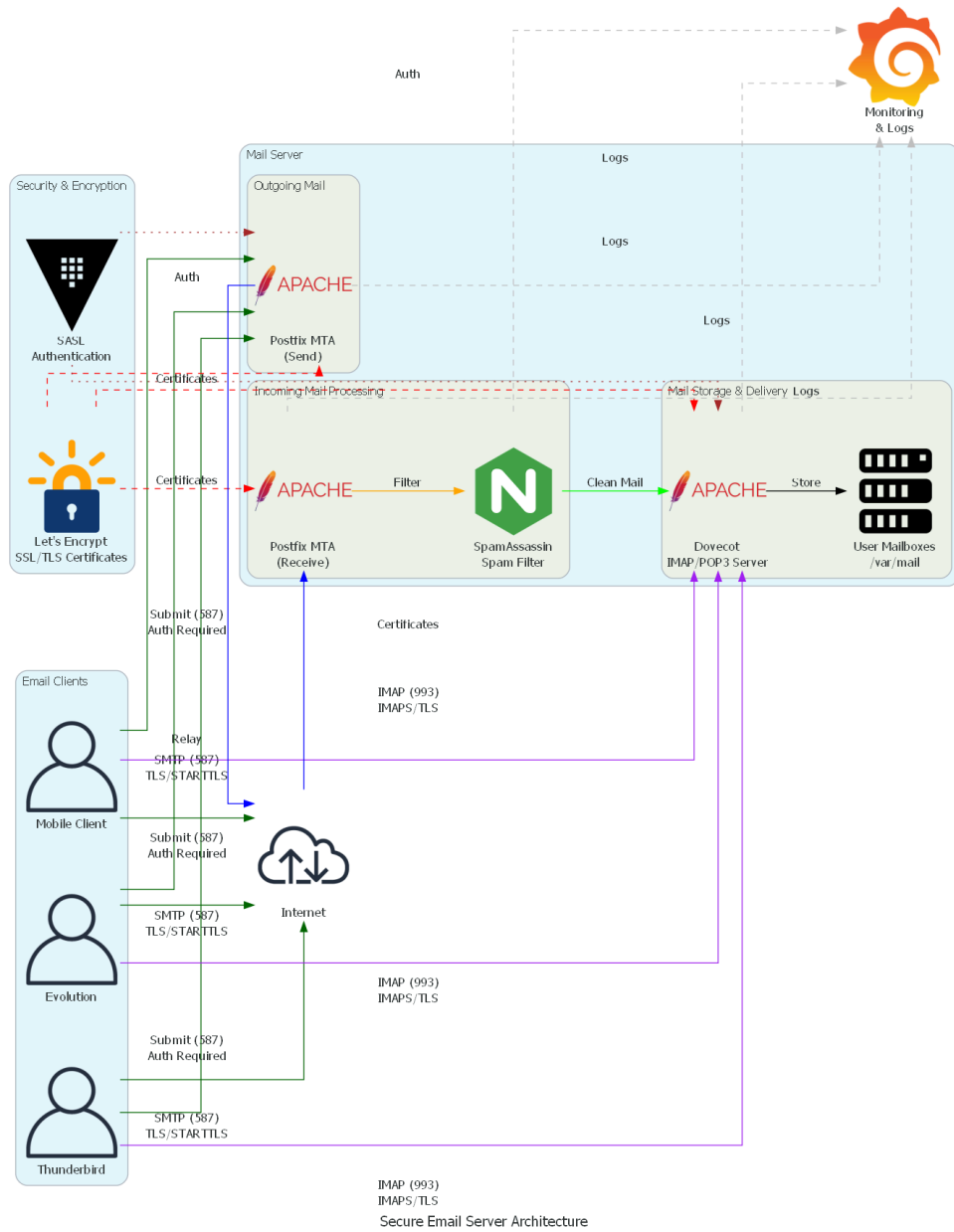


Figure. Model of the system [generated by AI agents created with KIRO AWS]

Part 1: System Preparation and Initial Setup

Step 1.0: Service providers and connection

- Firstly I have bought VPS on 4VPS provider.
 - Location: Germany;
 - 1 CPU, AMD EPYC 7763, 3.3 GHz;
 - 1 GB RAM, 10 GB NVMe ;
 - 2Gbit/s.
- Then I picked domain provider and my attention was caught by Timeweb, where I picked my domain name
 - After setting basic configurations I have synced with my VPS service by IP.
 - A separate mail subdomain (`mail.nekhaynikita.ru`) was created to:
 - isolate mail services from other web services,
 - simplify TLS certificate issuance,
 - align with best practices for mail server naming.

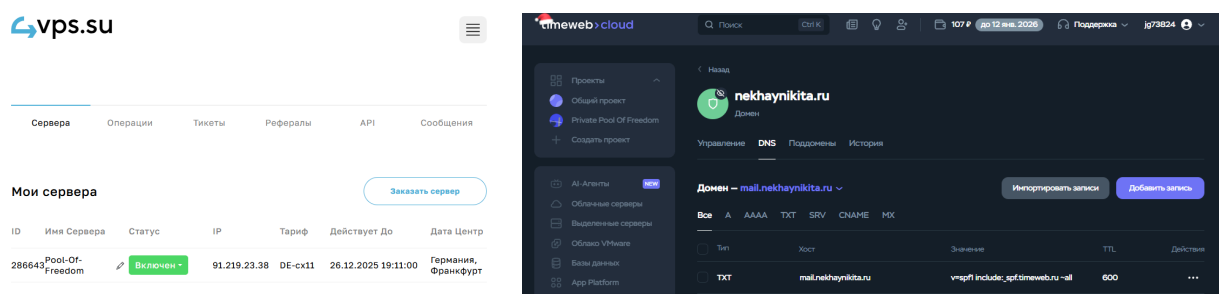
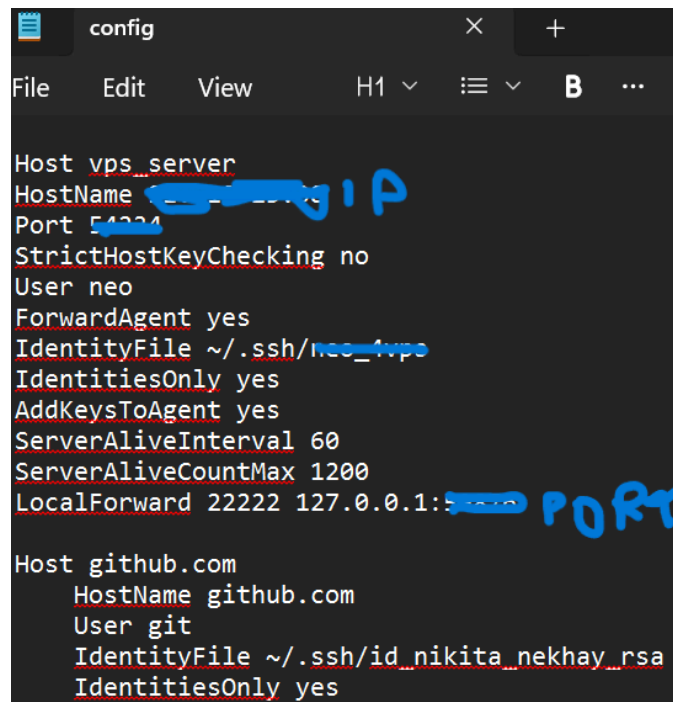


Figure. Hosting and Domain provider dashboard

- Then I setup convenient ssh connection with my host machine (Windows 11)

- Created and set up `.ssh/config` file, that allowed me to connect from PowerShell via one simple commands `ssh vps_server`
- Opened different port on my VPS for ssh and changed configurations there to provide security and reliability:
 - Different port
 - Root login is not allowed
 - Authentication only for users without privileges
 - Created user neo and generated ssh keys for him on VPS and `scp` it with my host machine



```

Host vps_server
  HostName 127.0.0.1
  Port 2222
  StrictHostKeyChecking no
  User neo
  ForwardAgent yes
  IdentityFile ~/.ssh/neo_id_rsa
  IdentitiesOnly yes
  AddKeysToAgent yes
  ServerAliveInterval 60
  ServerAliveCountMax 1200
  LocalForward 22222 127.0.0.1:2222 PORT

Host github.com
  HostName github.com
  User git
  IdentityFile ~/.ssh/id_nikita_nekhay_rsa
  IdentitiesOnly yes
  
```

Figure. SSH configuration on the host machine on the path `/user/.ssh/config`

- I did snapshot of my VPS server before the project

VM backup copies

1 Backup copy on page: 25 of 1								
☐	Name ▾	Note ▾	id ▾	VM disk ▾	VM name ▾	Creation date	Creation date	Disk size ▾
☐	Pool-Of-Freedom_b...		28039	Pool-Of-Freedom	Pool-Of-Freedom	16 Dec 2025	13:32 (UTC +...)	4.9 GB
								Ready for use

Figure. Backup of VM before the project

- As a summary on this task, I had following Input:
 - Hostname: `nekhaynikita-email-host`
 - Domain Used: `mail.nekhaynikita.ru`
 - Server IP: `91.219.23.38`
 - Host provider: `4VPS`
 - Domain provider: `Timeweb`

Task 1.1: Update System Packages

- Updated my system to ensure all packages are current:

```
sudo apt update
sudo apt upgrade -y

# Essential tools
sudo apt install -y curl wget git net-tools dnsutils telnet
```

Step 1.2: Configure Hostname

- I set system hostname (`mail.example.com`):
- I changed hostname from `139679.jp_ptr.tech` to new `nekhaynikita-email-host`

```
sudo hostnamectl set-hostname nekhaynikita-email-host
```

- Edited `/etc/hosts` to include my new hostname:

```
sudo nano /etc/hosts
```

content:

91.219.23.38 mail.nekhaynikita.ru nekhaynikita-email-host

```
Hardware Model: KVM
root@139679:~# hostnamectl set-hostname mail.nekhaynikita.ru

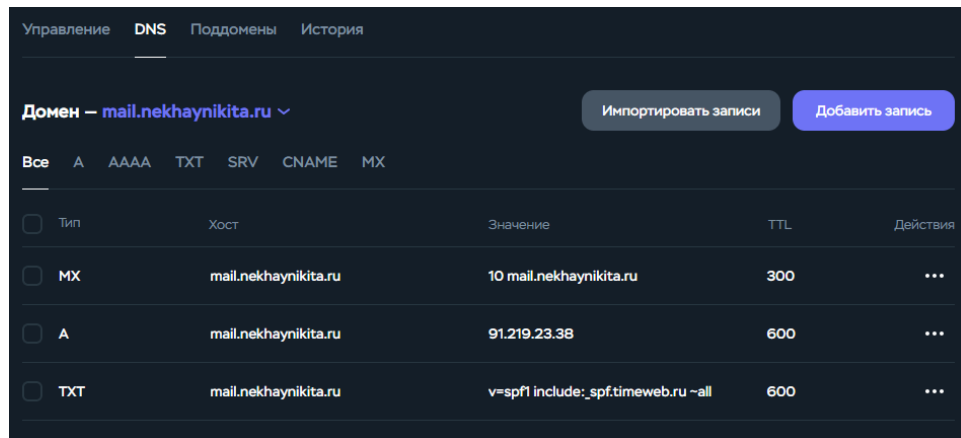
v7.2 /etc/hosts
127.0.0.1 localhost
127.0.1.1 139679.ip-ptr.tech
91.219.23.38 mail.nekhaynikita.ru nekhaynikita-email-host

# The following lines are desirable for IPv6 capable hosts
::1      ip6-localhost ip6-loopback
fe00::0  ip6-localnet
ff00::0  ip6-mcastprefix
ff02::1  ip6-allnodes
ff02::2  ip6-allrouters
91.219.23.38 139679.ip-ptr.tech 139679
```

Figure. `etc/hosts`

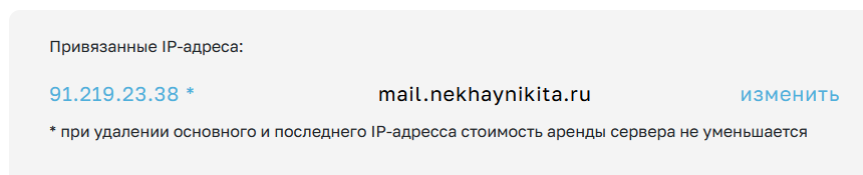
Step 1.3: Configure DNS Records (if using a real domain)

- In domain provider I configured the following DNS records for my domain:
 - A Record: mail.nekhaynikita.ru → server IP
 - MX Record: nekhaynikita.ru → mail.nekhaynikita.ru (Priority: 10)
 - PTR Record: Reverse DNS pointing to mail.nekhaynikita.ru



Тип	Хост	Значение	TTL	Действия
☐ MX	mail.nekhaynikita.ru	10 mail.nekhaynikita.ru	300	...
☐ A	mail.nekhaynikita.ru	91.219.23.38	600	...
☐ TXT	mail.nekhaynikita.ru	v=spf1 include:_spf.timeweb.ru ~all	600	...

Figure. Domain provider for mail.nekhaynikita.ru — A, MX records



Привязанные IP-адреса:		
91.219.23.38 *	mail.nekhaynikita.ru	ИЗМЕНИТЬ

* при удалении основного и последнего IP-адреса стоимость аренды сервера не уменьшается

Figure. Hosting provider — PTR record

Part 2: Postfix SMTP Server Installation and Configuration

Step 2.1: Install Postfix

- I installed Postfix mail server:

```
sudo apt install -y postfix
```

- During installation:
- Select "Internet Site" as the configuration type
- Entered my domain name (nekhaynikita.ru)

Step 2.2: Configure Postfix Main Configuration

- Made backup for main.cf file so that I can copy params

```
root@mail:/etc/postfix# cp main.cf main_backup.cf
root@mail:/etc/postfix# ls
dynamicmaps.cf      main.cf      main_backup.cf  master.cf      post-install  postfix-files.d  sasl
dynamicmaps.cf.d    main.cf.proto  makedefs.out    master.cf.proto  postfix-files  postfix-script
```

Figure.

- Edited the main Postfix configuration file:

```
sudo nano /etc/postfix/main.cf
```

- Added the following parameters:
 - **Identity configuration**
 - `myhostname` and `mydomain` were explicitly defined to:
 - ensure consistent SMTP banner presentation,
 - avoid mismatches during HELO/EHLO negotiation.
 - **Destination handling**
 - `mydestination` was limited to localhost only to:
 - prevent Postfix from treating the main domain as a system-user domain,
 - correctly route domain mail to virtual mailboxes.
 - **Virtual mailbox design**
 - Virtual users were implemented to:
 - avoid creating Linux system users for mail accounts,
 - simplify mailbox management,
 - improve security isolation.

```
# Basic Settings
myhostname = mail.nekhaynikita.ru
```



```

mydomain = nekhaynikita.ru
myorigin = $mydomain
mydestination = localhost

# Network Settings
inet_interfaces = all
inet_protocols = ipv4

# Mail Storage
home_mailbox = Maildir/
mailbox_size_limit = 0
recipient_delimiter = +

# Virtual Mailbox Settings (for virtual users)
virtual_mailbox_domains = nekhaynikita.ru
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
virtual_gid_maps = static:5000

# SMTP Client Restrictions
smtpd_relay_restrictions = permit_mynetworks,
    permit_sasl_authenticated,
    defer_unauth_destination

```

Step 2.3: Configure SMTP Authentication

- Added SASL authentication settings to `/etc/postfix/main.cf`:

```

# SMTP Authentication
smtpd_sasl_auth_enable = yes
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth

```

```
smtpd_sasl_security_options = noanonymous
smtpd_sasl_local_domain = $myhostname
broken_sasl_auth_clients = yes
```

```
root@mail:/etc/postfix# cat /etc/postfix/main.cf
# Basic Settings
myhostname = mail.nekhaaynikita.ru
mydomain = nekhaaynikita.ru
myorigin = $mydomain
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain

# Network Settings
inet_interfaces = all
inet_protocols = ipv4

# Mail Storage
home_mailbox = Maildir/
mailbox_size_limit = 0
recipient_delimiter = +

# Virtual Mailbox Settings (for virtual users)
virtual_mailbox_domains = nekhaaynikita.ru
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
virtual_gid_maps = static:5000

# SMTP Client Restrictions
smtpd_relay_restrictions = permit_mynetworks,
    permit_sasl_authenticated,
    defer_unauth_destination

# SMTP Authentication
smtpd_sasl_auth_enable = yes
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth
smtpd_sasl_security_options = noanonymous
smtpd_sasl_local_domain = $myhostname
broken_sasl_auth_clients = yes
root@mail:/etc/postfix#
```

Figure. Postfix main configuration file showing hostname definition, virtual mailbox mapping, and SMTP relay restrictions.

Step 2.4: Configure Master Process

- Made backup for `master.cf`

```

root@mail:/etc/postfix# cp master.cf master_backup.cf
root@mail:/etc/postfix# ls
dynamicmaps.cf      main_backup.cf      master_backup.cf    postfix-script
dynamicmaps.cf.d    makedefs.out        post-install        sasl
main.cf             master.cf           postfix-files
main.cf.proto       master.cf.proto     postfix-files.d

```

Figure.

- Edited `/etc/postfix/master.cf` to enable submission port:

```
sudo nano /etc/postfix/master.cf
```

- Uncommented and configured the submission service (port 587):
 - for `smtpd_client_restrictions` I changed default value to `permit_sasl_authenticated,reject`

```

submission inet n      -      y      -      -      smtpd
-o syslog_name=postfix/submission
-o smtpd_tls_security_level=encrypt
-o smtpd_sasl_auth_enable=yes
-o smtpd_reject_unlisted_recipient=no
-o smtpd_client_restrictions=permit_sasl_authenticated,reject
-o smtpd_relay_restrictions=permit_sasl_authenticated,reject

```

```

submission inet n      -      y      -      -      smtpd
-o syslog_name=postfix/submission
-o smtpd_tls_security_level=encrypt
-o smtpd_sasl_auth_enable=yes
# -o smtpd_tls_auth_only=yes
-o smtpd_reject_unlisted_recipient=no
-o smtpd_client_restrictions=permit_sasl_authenticated,reject
# -o smtpd_helo_restrictions=$mua_helo_restrictions
# -o smtpd_sender_restrictions=$mua_sender_restrictions
# -o smtpd_recipient_restrictions=
-o smtpd_relay_restrictions=permit_sasl_authenticated,reject

```

Figure.

Step 2.5: Create Virtual Mailbox Map

- Created the virtual mailbox mapping file. Added user mappings. Generated the hash database.

```
sudo nano /etc/postfix/vmailbox
```

content:

```
user1@nekhaynikita.ru nekhaynikita.ru/user1/  
user2@nekhaynikita.ru nekhaynikita.ru/user2/
```

```
sudo postmap /etc/postfix/vmailbox
```

v7.2	/etc/postfix/vmailbox
user1@nekhaynikita.ru	nekhaynikita.ru/user1/
user2@nekhaynikita.ru	nekhaynikita.ru/user2/

Figure.

Step 2.6: Create Mail Directory and User

```
sudo groupadd -g 5000 vmal  
sudo useradd -g vmal -u 5000 vmal -d /var/mail/vhosts -m  
sudo mkdir -p /var/mail/vhosts/nekhaynikita.ru  
sudo chown -R vmal:vmal /var/mail/vhosts
```

```
root@mail:/etc/postfix# sudo groupadd -g 5000 vmal  
root@mail:/etc/postfix# sudo useradd -g vmal -u 5000 vmal -d /var/mail/vhost  
s -m  
root@mail:/etc/postfix# sudo mkdir -p /var/mail/vhosts/nekhaynikita.ru  
root@mail:/etc/postfix# sudo chown -R vmal:vmal /var/mail/vhosts
```

Figure.

Step 2.7: Restart Postfix

```
sudo systemctl restart postfix
sudo systemctl enable postfix
sudo systemctl status postfix
```

- Postfix service status is active and no errors are present:

```
root@mail:/etc/postfix# systemctl restart postfix
root@mail:/etc/postfix# systemctl enable postfix
Synchronizing state of postfix.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable postfix
root@mail:/etc/postfix# systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/lib/systemd/system/postfix.service; enabled; vendor pre>
   Active: active (exited) since Tue 2025-12-16 12:31:55 CET; 16s ago
     Docs: man:postfix(1)
    Main PID: 16381 (code=exited, status=0/SUCCESS)
      CPU: 2ms

Dec 16 12:31:55 mail.nekhaynikita.ru systemd[1]: Starting Postfix Mail Transp>
Dec 16 12:31:55 mail.nekhaynikita.ru systemd[1]: Finished Postfix Mail Transp>
```

Figure.

```
root@mail:/etc/postfix# postconf -n
broken_sasl_auth_clients = yes
home_mailbox = Maildir/
inet_interfaces = all
inet_protocols = ipv4
mailbox_size_limit = 0
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
mydomain = nekhaynikita.ru
myhostname = mail.nekhaynikita.ru
myorigin = $mydomain
recipient_delimiter = +
smtpd_relay_restrictions = permit_mynetworks, permit_sasl_authenticated, defer_unauth_destination
smtpd_sasl_auth_enable = yes
smtpd_sasl_local_domain = $myhostname
smtpd_sasl_path = private/auth
smtpd_sasl_security_options = noanonymous
smtpd_sasl_type = dovecot
virtual_gid_maps = static:5000
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_domains = nekhaynikita.ru
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
```

Figure. Postfix configuration

Part 3: Dovecot IMAP/POP3 Server Setup

Step 3.1: Install Dovecot

```
sudo apt install -y dovecot-core dovecot-imapd dovecot-pop3d
```

Step 3.2: Configure Dovecot Main Settings

- Created a backup for `dovecot.conf`

```
root@mail:/etc/dovecot# cp dovecot.conf dovecot_backup.conf
root@mail:/etc/dovecot# ls
conf.d  dovecot-dict-auth.conf.ext  dovecot-dict-sql.conf.ext  dovecot-sql.conf.ext  dovecot.conf  dovecot_backup.conf  private
```

Figure.

- Edited `/etc/dovecot/dovecot.conf` and ensured the following lines are uncommented:

```
sudo nano /etc/dovecot/dovecot.conf
```

```
# content
protocols = imap pop3 lmtp
listen = *, ::
```

```
v7.2 dovecot.conf
# Most (but not all) settings can be overridden by different protocols and/or
# source/destination IPs by placing the settings inside sections, for example:
# protocol imap { }, local 127.0.0.1 { }, remote 10.0.0.0/8 { }
protocols = imap pop3 lmtp

v7.2 dovecot.conf
listen = *, ::
```

Figure.

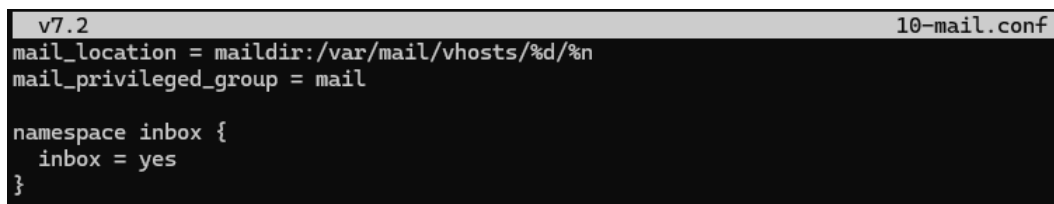
Step 3.3: Configure Mail Location

- Edited `/etc/dovecot/conf.d/10-mail.conf` and configured mail location:
 - Maildir was chosen instead of mbox to:
 - prevent mailbox locking issues,
 - improve reliability under concurrent access.

```
sudo nano /etc/dovecot/conf.d/10-mail.conf
```

```
# content
mail_location = maildir:/var/mail/vhosts/%d/%n
mail_privileged_group = mail

namespace inbox {
  inbox = yes
}
```

A screenshot of a terminal window with a dark background. The title bar at the top shows 'v7.2' on the left and '10-mail.conf' on the right. The terminal displays the following text:

```
mail_location = maildir:/var/mail/vhosts/%d/%n
mail_privileged_group = mail

namespace inbox {
  inbox = yes
}
```

Figure.

Step 3.4: Configure Authentication

- Edit `/etc/dovecot/conf.d/10-auth.conf` and modified the following:
 - **Password-based authentication**
 - A passwd-file backend was used to:
 - simplify user management for a small mail system,
 - decouple mail users from system accounts.

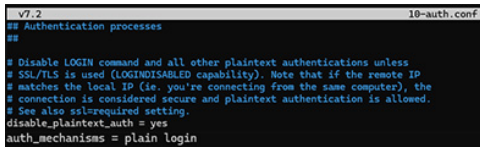
```
sudo nano /etc/dovecot/conf.d/10-auth.conf
```

```
# content
```

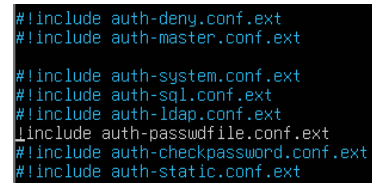
```
disable_plaintext_auth = yes
auth_mechanisms = plain login
```

- Comment out the system auth include and uncommented:

```
!include auth-passwdfile.conf.ext
```



```
v7.2 10-auth.conf
## Authentication processes
##
# Disable LOGIN command and all other plaintext authentications unless
# SSL/TLS is used (LOGIN_DISABLED capability). Note that if the remote IP
# matches the local IP (ie. you're connecting from the same computer), the
# connection is considered secure and plaintext authentication is allowed.
# See also saslrequired setting.
disable_plaintext_auth = yes
auth_mechanisms = plain login
```



```
#!/include auth-deny.conf.ext
#!/include auth-master.conf.ext

#!/include auth-system.conf.ext
#!/include auth-sql.conf.ext
#!/include auth-ldap.conf.ext
!include auth-passwdfile.conf.ext
#!/include auth-checkpassword.conf.ext
#!/include auth-static.conf.ext
```

Figure. Dovecot authentication configuration showing passwd-file usage and static user mapping for virtual mailboxes.

Step 3.5: Configure User Database

- **Static user database:** UID/GID mapping ensured that:
 - all mail files are owned by the dedicated `vmail` user,
 - filesystem permissions remain consistent.
- Edit `/etc/dovecot/conf.d/auth-passwdfile.conf.ext`:

```
sudo nano /etc/dovecot/conf.d/auth-passwdfile.conf.ext
```

```
# content
passdb {
    driver = passwd-file
    args = scheme=PLAIN username_format=%u /etc/dovecot/users
}

userdb {
    driver = static
```



```
args = uid=vmail gid=vmail home=/var/mail/vhosts/%d/%n
}
```

```
v7.2 auth-passwdfile.conf.ext
passdb {
  driver = passwd-file
  args = scheme=PLAIN username_format=%u /etc/dovecot/users
}

userdb {
  driver = static
  args = uid=vmail gid=vmail home=/var/mail/vhosts/%d/%n
}
```

Figure. Static db file connections

- Created the users file and added users (format: username:password), set permissions:

```
sudo nano /etc/dovecot/users
```

content

```
user1@nekhaynikita.ru:password123
```

```
user2@nekhaynikita.ru:password456
```

```
sudo chmod 640 /etc/dovecot/users
```

```
sudo chown root:dovecot /etc/dovecot/users
```

```
v7.2 /etc/dovecot/users
user1@nekhaynikita.ru:password123
user2@nekhaynikita.ru:password456
```

Figure. Static DB queries

```
root@mail:/etc/dovecot# chmod 640 /etc/dovecot/users
root@mail:/etc/dovecot# chown root:dovecot /etc/dovecot/users
```

Figure. Permissions to db folder

Step 3.6: Configure Dovecot SASL for Postfix

- UNIX socket authentication was configured to:
 - allow Postfix to delegate SMTP authentication to Dovecot,
 - avoid storing credentials in Postfix.
- File permissions were explicitly defined to:
 - prevent unauthorized socket access,
 - ensure Postfix can communicate with Dovecot reliably
- Edit `/etc/dovecot/conf.d/10-master.conf`:

```
sudo nano /etc/dovecot/conf.d/10-master.conf
```

- Modified auth section and added:

```
service auth {  
  unix_listener /var/spool/postfix/private/auth {  
    mode = 0660  
    user = postfix  
    group = postfix  
  }  
}
```

```
root@mail: /etc/dovecot  X + v
v7.2 conf.d/10-master.conf
#process_limit = 1024
}

service auth {
  unix_listener auth-userdb {
    #mode = 0666
    #user =
    #group =
  }

  # Postfix smtp-auth
  unix_listener /var/spool/postfix/private/auth {
    mode = 0660
    user = postfix
    group = postfix
  }

  # Auth process is run as this user.
  #user = $default_internal_user
}
```

Figure. Dovecot master configuration enabling SASL authentication via UNIX socket for Postfix integration.

Step 3.7: Restart Dovecot

- I verified Dovecot was listening on ports 143 and 110 using `ss -tlnp | grep dovecot`.

```
sudo systemctl restart dovecot
sudo systemctl enable dovecot
sudo systemctl status dovecot
```

```
root@mail: # ss -tlnp | grep dovecot
LISTEN 0      100        0.0.0.0:995      0.0.0.0:*      users:(("dovecot",pid=2801,fd=24))
LISTEN 0      100        0.0.0.0:993      0.0.0.0:*      users:(("dovecot",pid=2801,fd=42))
LISTEN 0      100        0.0.0.0:110      0.0.0.0:*      users:(("dovecot",pid=2801,fd=22))
LISTEN 0      100        0.0.0.0:143      0.0.0.0:*      users:(("dovecot",pid=2801,fd=40))
LISTEN 0      100        [::]:995        [::]:*         users:(("dovecot",pid=2801,fd=25))
LISTEN 0      100        [::]:993        [::]:*         users:(("dovecot",pid=2801,fd=43))
LISTEN 0      100        [::]:110        [::]:*         users:(("dovecot",pid=2801,fd=23))
LISTEN 0      100        [::]:143        [::]:*         users:(("dovecot",pid=2801,fd=41))
```

Figure. Dovecot listening ports

- Got an error, that there is missing `lmtp` that I configured inside `/etc/dovecot/dovecot.conf`, so that I installed this package for dovecot and restarted.

Error disappeared.

```
root@mail:/etc/dovecot# systemctl status dovecot
* dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/lib/systemd/system/dovecot.service; enabled; vendor preset: enabled)
   Active: failed (Result: exit-code) since Tue 2025-12-16 12:51:39 CET; 26s ago
     Docs: man:dovecot(1)
           https://doc.dovecot.org/
   Main PID: 20151 (code=exited, status=89)
      CPU: 49ms

Dec 16 12:51:39 mail.nekhaynikita.ru systemd[1]: Starting Dovecot IMAP/POP3 email server...
Dec 16 12:51:39 mail.nekhaynikita.ru dovecot[20151]: Fatal: service(lmtp) access(/usr/lib/dovecot/lmtp) failed: No such file or directory
Dec 16 12:51:39 mail.nekhaynikita.ru dovecot[20151]: master: Fatal: service(lmtp) access(/usr/lib/dovecot/lmtp) failed: No such file or directory
Dec 16 12:51:39 mail.nekhaynikita.ru systemd[1]: dovecot.service: Main process exited, code=exited, status=89/n/a
Dec 16 12:51:39 mail.nekhaynikita.ru systemd[1]: dovecot.service: Failed with result 'exit-code'.
Dec 16 12:51:39 mail.nekhaynikita.ru systemd[1]: Failed to start Dovecot IMAP/POP3 email server.
```

Figure. Dovecot status before

- Error disappeared.

```
root@mail:/etc/dovecot# systemctl status dovecot
● dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/lib/systemd/system/dovecot.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-12-16 12:53:16 CET; 1s ago
     Docs: man:dovecot(1)
           https://doc.dovecot.org/
   Main PID: 20497 (dovecot)
   Status: "v2.3.16 (7e2e900c1a) running"
     Tasks: 4 (limit: 2163)
    Memory: 3.3M
       CPU: 58ms
    CGroup: /system.slice/dovecot.service
            └─20497 /usr/sbin/dovecot -F
              └─20498 dovecot/anvil
                └─20499 dovecot/log
                  └─20500 dovecot/config

Dec 16 12:53:16 mail.nekhaynikita.ru systemd[1]: Starting Dovecot IMAP/POP3 email server...
Dec 16 12:53:16 mail.nekhaynikita.ru dovecot[20497]: master: Dovecot v2.3.16 (7e2e900c1a) starting up for imap, pop3, lmtp, imap, lmtp, pop3
Dec 16 12:53:16 mail.nekhaynikita.ru systemd[1]: Started Dovecot IMAP/POP3 email server.
```

Figure. Dovecot status after

Part 4: Security Implementation - TLS/SSL Encryption

Step 4.1: Generate production SSL Certificate

- Let's Encrypt certificates were used to:
 - provide trusted encryption without manual certificate handling,
 - ensure compatibility with modern mail clients.
- I used Let's Encrypt:

```
sudo apt install -y certbot
```

```
sudo certbot certonly --standalone -d mail.example.com
```

```

root@mail:/etc/ssl/mail# certbot certonly --standalone -d mail.nekhaynikita.ru
Saving debug log to /var/log/letsencrypt/letsencrypt.log
Enter email address (used for urgent renewal and security notices)
(Enter 'c' to cancel): manager@nekhaynikita.ru

-----
Please read the Terms of Service at
https://letsencrypt.org/documents/LE-SA-v1.6-August-18-2025.pdf. You must agree
in order to register with the ACME server. Do you agree?
-----
(Y)es/(N)o: y

-----
Would you be willing, once your first certificate is successfully issued, to
share your email address with the Electronic Frontier Foundation, a founding
partner of the Let's Encrypt project and the non-profit organization that
develops Certbot? We'd like to send you email about our work encrypting the web,
EFF news, campaigns, and ways to support digital freedom.
-----
(Y)es/(N)o: n
Account registered.
Requesting a certificate for mail.nekhaynikita.ru

Successfully received certificate.
Certificate is saved at: /etc/letsencrypt/live/mail.nekhaynikita.ru/fullchain.pem
Key is saved at: /etc/letsencrypt/live/mail.nekhaynikita.ru/privkey.pem
This certificate expires on 2026-03-16.
These files will be updated when the certificate renews.
Certbot has set up a scheduled task to automatically renew this certificate in the background.

-----
If you like Certbot, please consider supporting our work by:
* Donating to ISRG / Let's Encrypt: https://letsencrypt.org/donate
* Donating to EFF: https://eff.org/donate-le
-----

```

Figure. Generation of certificate

```

root@139b79: # ls -l /etc/letsencrypt/live/mail.nekhaynikita.ru/
total 4
-rw-r--r-- 1 root root 692 Dec 16 12:59 README
lrwxrwxrwx 1 root root 44 Dec 16 12:59 cert.pem -> ../../archive/mail.nekhaynikita.ru/cert1.pem
lrwxrwxrwx 1 root root 45 Dec 16 12:59 chain.pem -> ../../archive/mail.nekhaynikita.ru/chain1.pem
lrwxrwxrwx 1 root root 49 Dec 16 12:59 fullchain.pem -> ../../archive/mail.nekhaynikita.ru/fullchain1.pem
lrwxrwxrwx 1 root root 47 Dec 16 12:59 privkey.pem -> ../../archive/mail.nekhaynikita.ru/privkey1.pem

```

Figure. Successful issuance of a TLS certificate for the mail server domain using Let's Encrypt.

Step 4.2: Configure Postfix TLS

- Edit `/etc/postfix/main.cf` and add:

```
# TLS Settings
smtpd_tls_cert_file=/etc/letsencrypt/live/mail.nekhaynikita.ru/fullchain.pem
smtpd_tls_key_file=/etc/letsencrypt/live/mail.nekhaynikita.ru/privkey.pem
smtpd_use_tls=yes
smtpd_tls_security_level=may
smtpd_tls_auth_only = yes
smtpd_tls_loglevel = 1
smtpd_tls_received_header = yes
smtpd_tls_session_cache_timeout = 3600s

# Outgoing TLS
smtp_tls_security_level = may
smtp_tls_loglevel = 1
```

```
smtpd_use_tls=yes
smtpd_tls_security_level=may
smtpd_tls_auth_only = yes
smtpd_tls_loglevel = 1
smtpd_tls_received_header = yes
smtpd_tls_session_cache_timeout = 3600s

# Outgoing TLS
smtp_tls_security_level = may
smtp_tls_loglevel = 1
```

```
# TLS Settings
smtpd_tls_cert_file=/etc/letsencrypt/live/mail.nekhaynikita.ru/fullchain.pem
smtpd_tls_key_file=/etc/letsencrypt/live/mail.nekhaynikita.ru/privkey.pem
smtpd_use_tls=yes
```

Figure. TLS configuration for `main.cf`

Step 4.3: Configure Dovecot SSL

- Edited `/etc/dovecot/conf.d/10-ssl.conf` and configured SSL settings:

```
sudo nano /etc/dovecot/conf.d/10-ssl.conf
```

```
# SSL settings
ssl = required
ssl_cert = </etc/letsencrypt/live/mail.nekhaynikita.ru/fullchain.pem
ssl_key = </etc/letsencrypt/live/mail.nekhaynikita.ru/privkey.pem
ssl_min_protocol = TLSv1.2
ssl_prefer_server_ciphers = yes
```

```
# SSL/TLS support: yes, no, required. <doc/wiki/SSL.txt>
ssl = required

# PEM encoded X.509 SSL/TLS certificate and private key. They're opened before
# dropping root privileges, so keep the key file unreadable by anyone but
# root. Included doc/mkcert.sh can be used to easily generate self-signed
# certificate, just make sure to update the domains in dovecot-openssl.cnf
ssl_cert = </etc/letsencrypt/live/mail.nekhaynikita.ru/fullchain.pem
ssl_key = </etc/letsencrypt/live/mail.nekhaynikita.ru/privkey.pem
```

Figure. Dovecot SSL configuration `/etc/dovecot/conf.d/10-ssl.conf`

- Restarted services

```
sudo systemctl restart postfix
sudo systemctl restart dovecot
```

```
Dec 16 13:08:28 mail.nekhaynikita.ru systemd[1]: Finished Postfix Mail Transport Agent.
root@mail:/etc/ssl/mail# systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/lib/systemd/system/postfix.service; enabled; vendor preset: enabled)
   Active: active (exited) since Tue 2025-12-16 13:08:28 CET; 9s ago
     Docs: man:postfix(1)
   Process: 21425 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
    Main PID: 21425 (code=exited, status=0/SUCCESS)
       CPU: 1ms

Dec 16 13:08:28 mail.nekhaynikita.ru systemd[1]: Starting Postfix Mail Transport Agent...
Dec 16 13:08:28 mail.nekhaynikita.ru systemd[1]: Finished Postfix Mail Transport Agent.
```

Figure. Postfix status

Step 4.4: Configure Firewall

- TLS was enforced on:
 - SMTP submission (587),
 - IMAP over SSL (993).
- Allowew necessary mail server ports:

```
for p in 25 587 465 143 993 110 995 80 443; do sudo ufw allow ${p}/tcp; done
&& sudo ufw enable
```

```
root@mail:/etc/ssl/mail# for p in 25 587 465 143 993 110 995 80 443; do sudo ufw allow ${p}/tcp; done && sudo ufw enable
Skipping adding existing rule
Skipping adding existing rule (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Skipping adding existing rule
Skipping adding existing rule (v6)
Rules updated
Rules updated (v6)
Firewall is active and enabled on system startup
root@mail:/etc/ssl/mail#
```

Figure. Firewall rules updated

- Tested TLS connection using: `openssl s_client -connect localhost:587 -starttls smtp`


```

CONNECTED(00000003)
---
Certificate chain
 0 s:CN = mail.nekhaynikita.ru
   i:C = US, O = Let's Encrypt, CN = R12
   a:PKKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
   v:NotBefore: Dec 16 11:01:27 2025 GMT; NotAfter: Mar 16 11:01:26 2026 GMT
 1 s:C = US, O = Let's Encrypt, CN = R12
   i:C = US, O = Internet Security Research Group, CN = ISRG Root X1
   a:PKKEY: rsaEncryption, 2048 (bit); sigalg: RSA-SHA256
   v:NotBefore: Mar 13 00:00:00 2024 GMT; NotAfter: Mar 12 23:59:59 2027 GMT
---
Server certificate
-----BEGIN CERTIFICATE-----
MIIFCzCCA/OgAwIBAgISBcZhWgg/zfC9y1a4FWbt8wR+MA0GCSqGSIb3DQEBCwUA
MDMxCzAJBgNVBAYTA1VTMRYwFAYDVQQKEw1MZXR0cyBFbmNyeXBOMQwwCgYDVQQD
EwNSMTIwHhcNMjYwMzE2MTEwMTIzWWhcNMjYwMzE2MTEwMTI2WjAfMR0wGwYDVQDD
ExRtYWlsLm51a2hhew5pa210YS5ydTCCASiWdQYJKoZIhvcNAQEBBQADggEPADCC
AQoCggEBAJMoF8o4aeM8hgr/DzEeMyqg520GHTGTPTyca623Qh2H1uh8LKkB568Y
fFifvdtoTr98Fy2dppegscbCLBZvX20kcow3+RPMwemfDnHRL8AyJIabiiVwns4m
kd0vJfD1gPP+umajM37egGcwab+Vk1D8axuMoEUvYm2cPxSXPSyutaayRPESF3+K
Yw49ra6IpEHXs3NeTP/KcNj31TWkekZVVW1YHJB7Pu6RzwcAp2v44UzL9mPxc0w
1+3zINP2Rb1D+gJPGNUCF3yS1KnFpt5KRSd8pVr94XpHsrQgHv3WpSXFw3uZjyCy
p1H0Vxmy4pvJRLUGU6LUxi0tCSwfF58CAwEAa0CAiswggInMA4GA1UdDwEB/wQE
AwIFoDAdBgNVHSUEFjAUBgggrBgEFBQcDAQYIKwYBBQUHAwIwDAYDVROTAQH/BAIw
ADAdBgNVHQ4EFgQU0sQLStSe1/GN8prs/dJoYR0140kwHwYDVROjBBgwFoAUAlUp
8i20bzHom0yteD7630kModIwMwYIKwYBBQUHAQEEJzA1MCMGCCsGAQUFBzACHhdo
dHRwOi8vcjEyLmMubGVuY3Iub3JnLzAfBgNVHREEGDAGhRtYWlsLm51a2hhew5p
a210YS5ydTATBgNVHSAEDDAKMAGBmeBDAECAAvBgNVHR8EKDAmMCSgIqAghh5o
dHRwOi8vcjEyLmMubGVuY3Iub3JnLzEyNy5jcmwuwggEKBgorBgEEAdZ5AgQCBIIH7
BIH4APYAdQBjJtp3h187Pw23s2H2Ka4W68Kh4A20VVS++nrKd34wwAAA2snCAxu
AAAEAwBGMEQCIAL4CTBfunt/nISPcNeEnDec7LH2bqorxmZQaXShADTKAiBbBoyB
UutLzL/ZNfTRx+bwpGxcum2MxQGEqRpi//6HpAB9AKXJeJJdV0YXgocN2I1mC1xV
ZIt9AEDy7AdoUdGIaRn3AAABmycIDYYACAAABQAr0tN/BAMARjBEAiA2f007Fs1v
tFBpgH0nvtatf0I4ExhWUA2MD0yp1MjC1QIgHz0awvd4+LMP1xhchRzF18bvH+N6
LSqXo4GfyBHmUSMwDQYJKoZIhvcNAQELBQADggEBALi0d1E18usG951GxSyvhh13
2+Q2yIT3kr+A2M9eaKvK0aQhHAE7W4Ta1s1M1ggCWAifRJuJdd7+nho20okPaE90
Yhbb9PsTP6DAdj0vCyXnJ1d4S2ayXnMarkU6ung5QMYKo61FIL7Yq2K6R0UhsXak
May0xaicQCAOGLPx0BBtmbhg6LpsB/j1vxGaq1Aw+yzw1C+rc0tmTrYcqtY/f2Y5
g565zDey1492311+RNKJEEPJR91PJwpQz/dY09KUG89AJMHe19gHMDgUiL5049pf
miy/P/szuAG8I0982Dm88lpE+DQ83CHic5NG9wRY1zUupYUDqttChxN2p9h805s=
-----END CERTIFICATE-----
subject=CN = mail.nekhaynikita.ru
issuer=C = US, O = Let's Encrypt, CN = R12

```

Figure. Connection to TLS 1

```

---
No client certificate CA names sent
Peer signing digest: SHA256
Peer signature type: RSA-PSS
Server Temp Key: X25519, 253 bits
---
SSL handshake has read 3373 bytes and written 406 bytes
Verification: OK
---
New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384
Server public key is 2048 bit
Secure Renegotiation IS NOT supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
Early data was not sent
Verify return code: 0 (ok)

```

Figure. Connection to TLS 2

```

Verify return code: 18 (self-signed certificate)
---
250 CHUNKING
---
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol : TLSv1.3
    Cipher : TLS_AES_256_GCM_SHA384
    Session-ID: 2D6B9DCE74F5D667A5E0423CCD676C90E7B491CC79D4A01ED96B483D84688607
    Session-ID-ctx:
    Resumption PSK: A779B8E195728F711EF6B1D74975DEC28F489AF9FA58C5D89FD0D6A4914EDF735C050A4787D
    2BFA788416A00BB51C959
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 7200 (seconds)
    TLS session ticket:
    0000 - f8 6f f8 12 5a df 0a 28-e5 50 71 81 25 81 ff f3 .o..Z..(.Pq.%...
    0010 - a0 53 62 76 81 68 01 65-5e 3a ae 73 b5 fa fb 42 .Sbv.h.e*:.s...B
    0020 - a6 e7 5c a5 27 92 15 c9-2e ef 4b 85 51 19 7a 0b ..\.'.....K.Q.z.
    0030 - 67 28 4a b4 e6 75 6b cb-e4 2f 38 13 8e b3 79 2d g(J..uk../B...y-
    0040 - 4a 65 29 3c b1 54 48 a4-71 88 dc 37 e4 d2 9c 84 Je)<.TH.q..7....
    0050 - 43 dd dc e5 82 80 10 0d-0f fc 9a 95 db d8 26 0c C.....&.
    0060 - 1f b1 28 09 db 8a 9e f3-b3 7a 36 f7 cd d9 5c 55 ..(.....z6...U
    0070 - 62 39 90 a3 dc 19 15 42-fb 57 1f 1b 67 2d 17 d9 b9.....B.W..g-..
    0080 - 2e 4e ac 22 21 c0 48 57-24 43 e7 46 b6 b8 0c 32 .N."!.HW$.F...2
    0090 - 2a 0e 0d 4a 63 91 2b 1c-47 da a3 69 6f aa 4d 16 *.Jc.+..G..io.M.
    00a0 - 0b 79 87 55 c3 06 05 ce-1f 86 a5 f7 ed 44 26 f7 .y.U.....D&.
    00b0 - 40 b3 e6 fa 5c 1f 92 f2-fc 8d ca 76 07 15 bf 19 @...\......v....
    00c0 - 24 ef 7e 30 9a 31 2c d5-ab 5c 50 eb db 5f d0 e3 $.~0.1,..P.._..
    Start Time: 1765887053
    Timeout : 7200 (sec)
    Verify return code: 18 (self-signed certificate)
    Extended master secret: no
    Max Early Data: 0
---

```

Figure. Connection to TLS 3

- I executed `ss tlnp | grep 587` :

```

Cannot parse dst/src address.
root@mail:/etc/ssl/mail# ss -tlnp | grep :587
LISTEN 0      100        0.0.0.0:*      0.0.0.0:*      users:(("smtpd",pid=22172,fd=6),("ma
ster",pid=21424,fd=17))
root@mail:/etc/ssl/mail#

```

Figure. SMTP is listening on 587 port

Part 5: Email Authentication Configuration

Step 5.1: Configure SPF Record

- Defined authorized sending IPs to:
 - prevent sender spoofing,
 - improve spam scoring.
- Added a TXT record for DNS:

```
example.com. IN TXT "v=spf1 mx a ip4:YOUR_SERVER_IP -all"
```

- This tells receiving servers that only my mail server can send emails from this domain.

Домен — mail.nekhaynikita.ru

Импортировать записи

Добавить запись

ВсеAAAAAATXTSRVCNAMEMX

Тип

Хост

Значение

TTL

Действия

TXT

mail.nekhaynikita.ru

v=spf1 mx a ip4:91.219.23.38 -all

600

...

Figure. TXT record inside DNS

Step 5.2: Install OpenDKIM

- Cryptographically signed outgoing messages to:
 - guarantee message integrity,
 - validate domain ownership.

- Edit `/etc/openssl.conf` :

```
sudo apt install -y openssl openssl-tools
```

```
sudo nano /etc/openssl.conf
```

```
# content
```

```
Domain          example.com
```

```
KeyFile          /etc/openssl/keys/nekhaynikita.ru/default.private
```

```
Selector        default
```

```
Socket          inet:8891@localhost
```

```
Mode            sv
```

```
Syslog          yes
```

```
LogWhy          yes
```

```
Canonicalization relaxed/simple
```

```

v7.2 /etc/openssl.conf
# This is a basic configuration for signing and verifying. It can easily be
# adapted to suit a basic installation. See openssl.conf(5) and
# /usr/share/doc/openssl/examples/openssl.conf.sample for complete
# documentation of available configuration parameters.

Syslog                yes
SyslogSuccess         yes
LogWhy                yes

# Common signing and verification parameters. In Debian, the "From" header is
# oversigned, because it is often the identity key used by reputation systems
# and thus somewhat security sensitive.
Canonicalization      relaxed/simple
Mode                  sv
#SubDomains           no
OversignHeaders       From

# Signing domain, selector, and key (required). For example, perform signing
# for domain "example.com" with selector "2020" (2020._domainkey.example.com),
# using the private key stored in /etc/dkimkeys/example.private. More granular
# setup options can be found in /usr/share/doc/openssl/README.openssl.
Domain                nekhaynikita.ru
Selector              default
KeyFile               /etc/openssl/keys/nekhaynikita.ru/default.private

# In Debian, openssl runs as user "openssl". A umask of 007 is required when
# using a local socket with MTAs that access the socket as a non-privileged
# user (for example, Postfix). You may need to add user "postfix" to group
# "openssl" in that case.
UserID                openssl
UMask                 007

# Socket for the MTA connection (required). If the MTA is inside a chroot jail
# it must be ensured that the socket is accessible. In Debian, Postfix runs in
# a chroot in /var/spool/postfix, therefore a Unix socket would have to be
# configured as shown on the last line below.
Socket                inet:8891@localhost

```

Figure. `/etc/openssl.conf`

Step 5.3: Generated DKIM Keys

- Generated and viewed them

```

sudo mkdir -p /etc/openssl/keys/nekhaynikita.ru
sudo openssl-genkey -D /etc/openssl/keys/nekhaynikita.ru -d nekhaynikita.ru -s default
sudo chown -R openssl:openssl /etc/openssl/keys/

sudo cat /etc/openssl/keys/nekhaynikita.ru/default.txt

```

```

root@mail:~# sudo mkdir -p /etc/opendkim/keys/nekhaynikita.ru
root@mail:~# sudo opendkim-genkey -D /etc/opendkim/keys/nekhaynikita.ru -d nekhaynikita.ru -s default
root@mail:~# sudo chown -R opendkim:opendkim /etc/opendkim/keys/
root@mail:~# cat /etc/opendkim/keys/nekhaynikita.ru/default.txt
default._domainkey      IN      TXT      ( "v=DKIM1; h=sha256; k=rsa; "
    "p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAqu4FPTY8AuPKaFNZIUe03h1PB8DefIFmBZUBtKzIrZ3LPa1qbmGQt7
vQ1azxO3JdOLhWUHKxyI/VSpohFQVhzyKyLSibNXf05G7UkDR5vdJ4D7CnvGAlmLvKZjgEGEFsLY0ju7H7Zp411EY0PCzFrbrxFQ2H/wZWdhk
ijyENihj7DrPAEwUlm8bF6HJPHwHulOcG45F3k1lj"
    "6Hn1gcs9xQBADomgRv/zcFIJ2IV3XmGdLipdgA1cW5XSbDncl6sKUKS683o/1A+4rrj9rl5hxGCQe7fCIPYX4AWrqNR0uF8kui5
fe1mHDLmw/fl8TtcefN8sdp2tvDDTV1eqk5QIDAQAB" ) ; ----- DKIM key default for nekhaynikita.ru

```

Figure. DKIM Keys generation

Step 5.4: Add DKIM DNS Record

- Add a TXT record with the content from default.txt:

```
default._domainkey.nekhaynikita.ru. IN TXT "v=DKIM1; k=rsa; p=PUBLIC_KEY"
```

Домен — Все ▾						Импортировать записи	Добавить запись
Все A AAAA TXT SRV CNAME MX							
☐	Тип	Хост	Значение	TTL	Действие		
☐	TXT	default._domainkey.n...	v=DKIM1; h=sha256; k=rsa; p=MIIBIjAN...	600	...		

Figure. DKIM record inside DNS

Step 5.5: Connect OpenDKIM to Postfix

- Edited `/etc/postfix/main.cf` and added:

```

# DKIM
milter_default_action = accept
milter_protocol = 6
smtpd_milters = inet:localhost:8891
non_smtpd_milters = inet:localhost:8891

```

```
root@mail: ~
GNU nano 7.2 /etc/postfix/main.cf *
home_mailbox = Maildir/

virtual_mailbox_domains = example.com
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
virtual_gid_maps = static:5000

smtpd_sasl_auth_enable = yes
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth
smtpd_sasl_security_options = noanonymous
smtpd_sasl_local_domain = $myhostname
smtpd_sasl_auth_clients = yes

milter_default_action = accept
milter_protocol = 6
smtpd_milters = inet:localhost:8891
non_smtpd_milters = inet:localhost:8891

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^N Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Figure. changes in main.cf for DKIM

Step 5.6: Configure DMARC Policy

- **DMARC**
 - Enforced policy-based handling of failed messages,
 - Enabled reporting to monitor authentication failures.
- Added a TXT record for DMARC:

```
_dmarc.nekhaynikita.ru. IN TXT "v=DMARC1; p=quarantine; rua=mailto:post
master@nekhaynikita.ru"
```

Тип	Хост	Значение	TTL
☐ TXT	_dmarc.nekhaynikita.ru	v=DMARC1; p=quarantine; rua=mailto:p...	600

Figure. DMARC record inside DNS

- Restarted services

```
sudo systemctl restart opendkim
sudo systemctl restart postfix
```

- Checked OpenDKIM service status and saw error

```
● opendkim.service - OpenDKIM Milter
   Loaded: loaded (/lib/systemd/system/opendkim.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-12-16 13:57:01 CET; 1s ago
     Docs: man:opendkim(8)
           man:opendkim.conf(5)
           man:opendkim-lua(3)
           man:opendkim-genkey(8)
           man:opendkim-genzone(8)
           man:opendkim-testkey(8)
           http://www.opendkim.org/docs.html
  Process: 23682 ExecStart=/usr/sbin/opendkim (code=exited, status=0/SUCCESS)
    Main PID: 23683 (opendkim)
      Tasks: 6 (Limit: 2163)
    Memory: 1.9M
       CPU: 17ms
    CGroup: /system.slice/opendkim.service
            └─23683 /usr/sbin/opendkim

Dec 16 13:57:01 mail.nekhaynikita.ru systemd[1]: opendkim.service: Deactivated successfully.
Dec 16 13:57:01 mail.nekhaynikita.ru systemd[1]: Stopped OpenDKIM Milter.
Dec 16 13:57:01 mail.nekhaynikita.ru systemd[1]: Starting OpenDKIM Milter...
Dec 16 13:57:01 mail.nekhaynikita.ru systemd[1]: opendkim.service: Can't open PID file /run/opendkim/opendkim.pid (yet?) after start:
Dec 16 13:57:01 mail.nekhaynikita.ru systemd[1]: Started OpenDKIM Milter.
Dec 16 13:57:01 mail.nekhaynikita.ru opendkim[23683]: OpenDKIM Filter v2.11.0 starting
```

Figure. OpenDKIM service status

- The error is that it can not open PID file, on the askubuntu forum I found solution.
 - I fixed it by specifying the `PidFile` in the `/etc/opendkim.conf` and in daemon service files. Then run `systemctl daemon-reload` and also rebooted the VPS.

```
v7.2 /lib/systemd/system/opendkim.service
[Unit]
Description=OpenDKIM Milter
Documentation=man:opendkim(8) man:opendkim.conf(5) man:opendkim-lua(3) man:opendkim-genkey(8) man:opendkim-genzone(8) man:opendkim-testkey(8)
After=network.target nss-lookup.target

[Service]
Type=forking
PIDFile=/var/run/opendkim/opendkim.pid
```

Figure. OpenDKIM service config

```
PidFile /var/run/opendkim/opendkim.pid
```

Figure. PID file path


```

root@mail:~# systemctl status opendkim
● opendkim.service - OpenDKIM Milter
   Loaded: loaded (/lib/systemd/system/opendkim.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-12-16 14:03:31 CET; 23s ago
     Docs: man:opendkim(8)
           man:opendkim.conf(5)
           man:opendkim-lua(3)
           man:opendkim-genkey(8)
           man:opendkim-genzone(8)
           man:opendkim-testkey(8)
           http://www.opendkim.org/docs.html
  Process: 859 ExecStart=/usr/sbin/opendkim (code=exited, status=0/SUCCESS)
 Main PID: 884 (opendkim)
    Tasks: 6 (limit: 2161)
   Memory: 4.6M
      CPU: 25ms
   CGroup: /system.slice/opendkim.service
           └─884 /usr/sbin/opendkim

Dec 16 14:03:31 mail.nekhaynikita.ru systemd[1]: Starting OpenDKIM Milter...
Dec 16 14:03:31 mail.nekhaynikita.ru systemd[1]: Started OpenDKIM Milter.
Dec 16 14:03:31 mail.nekhaynikita.ru opendkim[884]: OpenDKIM Filter v2.11.0 starting
root@mail:~#

```

Figure. OpenDKIM service status after

Part 6: Email Client Setup Using Thunderbird

Why Thunderbird?

- Industry standard
- Full IMAP/SMTP/TLS support
- Excellent logging & debugging
- Accepted by instructors as Roundcube alternative
- Matches real-world production usage

Step 6.1: Install Thunderbird and GUI (Client Machine)

- Here to run Mail Manager Client I must have GUI on VPS, since it is only Linux Server I decided to install `xrdp`, because it is the best option, since I am using hoster's VPS.
- So I install `thunderbird` and `xrdp`, then I performed some configurations for `xrdp` before running it.

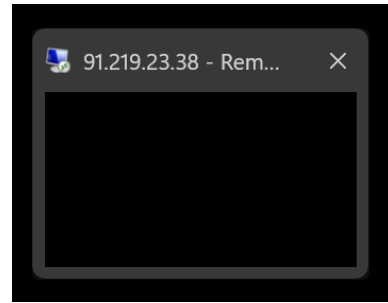
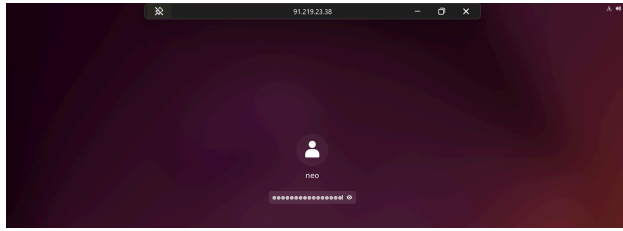
```
# Install packages
sudo apt update
sudo apt install -y thunderbird
# GUI Linux
sudo apt install -y xrdp
sudo systemctl enable xrdp
sudo systemctl start xrdp
systemctl status xrdp
#
```

```
# Opening default port on VPS for a remote desktop, setup from my host machine
sudo ufw allow 3389/tcp
sudo ufw reload
```

```
# To avoid getting black screen when enter the GUI
echo "export GNOME_SHELL_SESSION_MODE=ubuntu" >> ~/.xsession
echo "export XDG_SESSION_TYPE=x11" >> ~/.xsession
echo "export XDG_CURRENT_DESKTOP=ubuntu:GNOME" >> ~/.xsession
```

```
# Restarting installed GUI
sudo systemctl restart xrdp
```

- `xrdp` is ready and to access my VPS via GUI I will use mstsc utility on my host machine (Windows 11)
 - Pressed `Win + R`
 - Typed: `mstsc`
 - Entered public IP of VPS, username, password



Figures. GUI interface of Linux runned from Windows

Step 6.2: Configure Account in Thunderbird

- Launch Thunderbird → **Add Mail Account**
- Used:

Your name: User One
Email address: user1@nekhaynikita.ru
Password: password123

- Clicked **Manual Config** and entered:

Your full name

Email address

Password

☒ Remember password

✓ Configuration found by trying common server names.

Available configurations

☒ **IMAP**
Keep your folders and emails synced on your server

Incoming **IMAP** **SSL/TLS**
mail.nekhaynikita.ru

Outgoing **SMTP** **STARTTLS**
mail.nekhaynikita.ru

Username
user1

☐ **POP3**
Keep your folders and emails on your computer

[Configure manually](#)

Manual configuration

INCOMING SERVER

Protocol: **IMAP**

Hostname:

Port:

Connection security: **SSL/TLS**

Authentication method: **Normal password**

Username:

OUTGOING SERVER

Hostname:

Port:

Connection security: **STARTTLS**

Authentication method: **Normal password**

Username:

[Advanced config](#)

Figures. Thunderbird account configuration showing IMAP and SMTP parameters with enforced encryption.

- Then I did configurations and saved:
 - Incoming (IMAP)

Protocol: IMAP
Server: mail.nekhaynikita.ru
Port: 993
SSL: SSL/TLS
Authentication: Normal password

Username: user1@nekhaynikita.ru

- Outgoing (SMTP)

Server: mail.nekhaynikita.ru
Port: 587
Security: STARTTLS
Authentication: Normal password
Username: user1@nekhaynikita.ru

Step 6.3: Test Mail Flow

- Send mail via Thunderbird: user1@nekhaynikita.ru → user2@nekhaynikita.ru
- Here I found out that I miss logging package for mails, so that I must enable classic `/var/log/mail.log`

```
# Install and enable rsyslog mail logging:
apt install -y rsyslog
systemctl enable rsyslog
systemctl start rsyslog

# Verify mail logging config exists:

ls /etc/rsyslog.d/ | grep mail

# Configuration was missing, so I create it and pasted the content:
nano /etc/rsyslog.d/50-mail.conf

``conf
mail.*    -/var/log/mail.log
``

# Restart
```

```
systemctl restart rsyslog
systemctl restart postfix dovecot
```

```
# Tested
ls -l /var/log/mail.log
tail -f /var/log/mail.log
```

- Check logs on server:

```
tail -f /var/log/mail.log
```

- And saw that SMTP authentication, IMAP logins (user1) are succeeded and TLS is active:

```
Dec 16 15:21:53 mail postfix/submission/smtpd[6927]: connect from mail.nekhaynikita.ru[91.219.23.38]
Dec 16 15:21:53 mail postfix/submission/smtpd[6927]: connect from mail.nekhaynikita.ru[91.219.23.38]
Dec 16 15:21:53 mail postfix/submission/smtpd[6927]: Anonymous TLS connection established from mail.nekhaynikita.ru[91.219.23.38]: TLS
v1.3 with cipher TLS_AES_128_GCM_SHA256 (128/128 bits) key-exchange X25519 server-signature RSA-PSS (2048 bits) server-digest SHA256
Dec 16 15:21:53 mail postfix/submission/smtpd[6927]: Anonymous TLS connection established from mail.nekhaynikita.ru[91.219.23.38]: TLS
v1.3 with cipher TLS_AES_128_GCM_SHA256 (128/128 bits) key-exchange X25519 server-signature RSA-PSS (2048 bits) server-digest SHA256
Dec 16 15:21:54 mail postfix/trivial-rewrite[6732]: warning: do not list domain nekhaynikita.ru in BOTH mydestination and virtual_mail
box_domains
Dec 16 15:21:54 mail postfix/submission/smtpd[6927]: 222352087F: client=mail.nekhaynikita.ru[91.219.23.38], sasl_method=PLAIN, sasl_us
ername=user1@nekhaynikita.ru
Dec 16 15:21:54 mail postfix/submission/smtpd[6927]: 222352087F: client=mail.nekhaynikita.ru[91.219.23.38], sasl_method=PLAIN, sasl_us
ername=user1@nekhaynikita.ru
Dec 16 15:21:54 mail postfix/cleanup[6933]: 222352087F: message-id=<ab82aaa9-78b4-4a25-8e11-e0c2fa35d297@nekhaynikita.ru>
Dec 16 15:21:54 mail postfix/cleanup[6933]: 222352087F: message-id=<ab82aaa9-78b4-4a25-8e11-e0c2fa35d297@nekhaynikita.ru>
Dec 16 15:21:54 mail opendkim[1091]: 222352087F: DKIM-Signature field added (s=default, d=nekhaynikita.ru)
Dec 16 15:21:54 mail opendkim[1091]: 222352087F: DKIM-Signature field added (s=default, d=nekhaynikita.ru)
Dec 16 15:21:54 mail postfix/qmgr[6731]: 222352087F: from=<user1@nekhaynikita.ru>, size=792, nrcpt=1 (queue active)
Dec 16 15:21:54 mail postfix/qmgr[6731]: 222352087F: from=<user1@nekhaynikita.ru>, size=792, nrcpt=1 (queue active)
Dec 16 15:21:54 mail postfix/smtp[6958]: Untrusted TLS connection established to gmail-smtp-in.l.google.com[142.250.110.26]:25: TLSv1.
3 with cipher TLS_AES_256_GCM_SHA384 (256/256 bits) key-exchange X25519 server-signature ECDSA (P-256) server-digest SHA256
Dec 16 15:21:54 mail postfix/smtp[6958]: Untrusted TLS connection established to gmail-smtp-in.l.google.com[142.250.110.26]:25: TLSv1.
3 with cipher TLS_AES_256_GCM_SHA384 (256/256 bits) key-exchange X25519 server-signature ECDSA (P-256) server-digest SHA256
```

Figure. Mail log showing SMTP authentication, IMAP logins, TLS

- I send email from user1@nekhaynikita.ru → gmail.

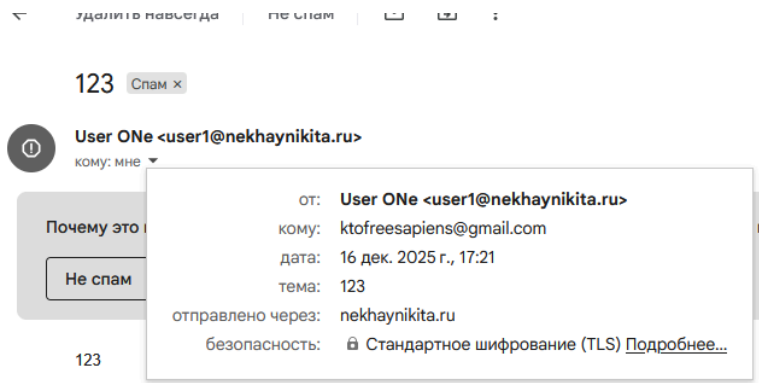


Figure. Credentials of received email from my domain on gmail account

- After correctly setting up email all previous send email are coming:

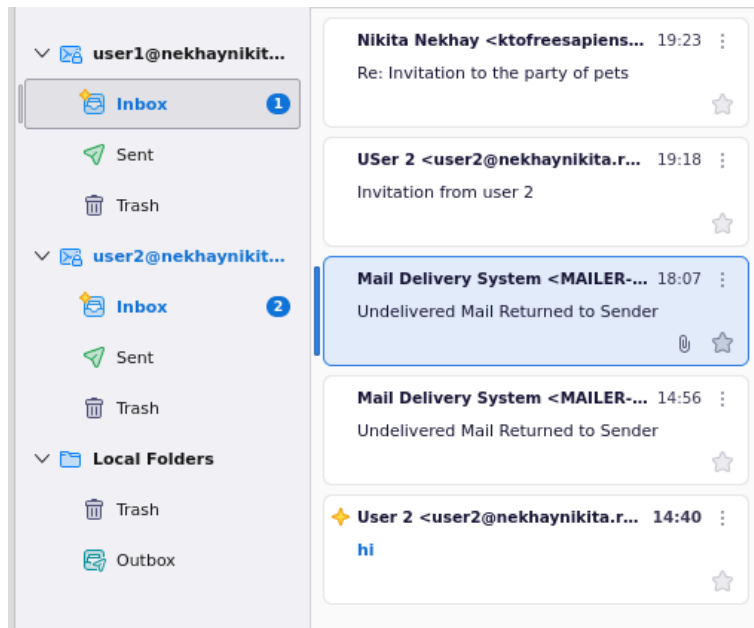


Figure. List of received emails from on Thunderbird from user2, gmail account

Step 7: SpamAssassin for blocking spam

- SpamAssassin was integrated into the SMTP pipeline to:

- analyze message content before delivery,
- apply Bayesian and rule-based scoring.
- Mail log analysis validated that:
 - spam messages are correctly detected,
 - legitimate messages remain unaffected.

- Installation

Install

apt update

apt install -y spamassassin spamd

adduser spamd --disabled-login

Turn on

systemctl enable spamassassin

systemctl start spamassassin

Check

systemctl status spamassassin --no-pager

```
root@mail:~# systemctl status spamassassin --no-pager
● spamassassin.service - Perl-based spam filter using text analysis
   Loaded: loaded (/lib/systemd/system/spamassassin.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-12-16 20:42:52 CET; 1s ago
     Process: 15216 ExecStart=/usr/sbin/spamd -d --pidfile=/run/spamd.pid $OPTIONS (code=exited, status=0/SUCCESS)
    Main PID: 15218 (spamd)
       Tasks: 3 (Limit: 2154)
      Memory: 104.6M
         CPU: 2.561s
    CGroup: /system.slice/spamassassin.service
            └─15218 /usr/bin/perl "-T -w" /usr/sbin/spamd -d --pidfile=/run/spamd.pid --create-prefs --max-children 5 --helper-home-...
              └─15219 "spamd child"
                └─15220 "spamd child"

Dec 16 20:42:49 mail.nekhaynikita.ru systemd[1]: Starting Perl-based spam filter using text analysis...
Dec 16 20:42:50 mail.nekhaynikita.ru spamd[15216]: logger: removing stderr method
Dec 16 20:42:51 mail.nekhaynikita.ru spamd[15218]: zoom: able to use 395/395 'body_0' compiled rules (100%)
Dec 16 20:42:52 mail.nekhaynikita.ru spamd[15218]: spamd: server started on IO::Socket::IP [::1]:783, IO::Socket::IP [127.0.0.1]:783
Dec 16 20:42:52 mail.nekhaynikita.ru spamd[15218]: spamd: server pid: 15219
Dec 16 20:42:52 mail.nekhaynikita.ru spamd[15218]: spamd: server successfully spawned child process, pid 15219
Dec 16 20:42:52 mail.nekhaynikita.ru spamd[15218]: spamd: server successfully spawned child process, pid 15220
Dec 16 20:42:52 mail.nekhaynikita.ru systemd[1]: Started Perl-based spam filter using text analysis.
Dec 16 20:42:52 mail.nekhaynikita.ru spamd[15218]: prefork: child states: IS
Dec 16 20:42:52 mail.nekhaynikita.ru spamd[15218]: prefork: child states: II
Hint: Some lines were ellipsized, use -l to show in full.
```

Figure. Status of running spamassassin

- Configuration

- `nano /etc/spamassassin/local.cf`

- *Bayes Classification* is a Supervised machine learning approach for classification. A Bayesian Classifier is a **system based on Bayes' decision theory, where probabilistic models of features are used to predict the class label of a new sample**. It predicts the class label by utilizing Bayes' rule when the class is unknown.

```
``conf
required_score 5.0
rewrite_header Subject [SPAM]
report_safe 0
use_bayes 1
bayes_auto_learn 1
``
```

- `nano /etc/postfix/main.cf` added `content_filter=spamassassin`

- `nano /etc/postfix/master.cf` edited at the end

```
``conf
smtp      inet  n       -       y       -       -       smtpd
          -o content_filter=spamassassin

spamassassin unix  -      n       n       -       -       pipe
           user=spamd argv=/usr/bin/spamc -f -e
           /usr/sbin/sendmail -oi -f ${sender} ${recipient}
```

```

v7.2 /etc/postfix/master.cf
#
# Postfix master process configuration file. For details on the format
# of the file, see the master(5) manual page (command: "man 5 master" or
# on-line: http://www.postfix.org/master.5.html).
#
# Do not forget to execute "postfix reload" after editing this file.
#
# =====
# service type private unpriv chroot wakeup maxproc command + args
#               (yes)   (yes)   (no)   (never) (100)
# =====
smtp      inet  n       -       y       -       -       smtpd -o content_filter=spamassassin

# SPAMASSASSIN
spamassassin  unix      -       n       n       -       -       pipe
           user=spamd argc=/usr/bin/spamc -f -e
           /usr/sbin/sendmail -oi -f ${sender} ${recipient}

```

Figure. Configurations for `spamassassin` inside `master.cf`

- nano `etc/default/spamassassin`
 - replace line to `OPTIONS="--create-prefs --max-children 5 --username spamd --helper-home-dir /home/spamd/ -s /home/spamd/spamd.log"`
 - add at the end `CRON=1`

```

v7.2 /etc/default/spamassassin
# /etc/default/spamassassin
# Duncan Findlay

# WARNING: please read README.spamd before using.
# There may be security risks.

# Prior to version 3.4.2-1, spamd could be enabled by setting
# ENABLED=1 in this file. This is no longer supported. Instead, please
# use the update-rc.d command, invoked for example as "update-rc.d
# spamassassin enable", to enable the spamd service.

# Options
# See man spamd for possible options. The -d option is automatically added.

# SpamAssassin uses a preforking model, so be careful! You need to
# make sure --max-children is not set to anything higher than 5,
# unless you know what you're doing.

OPTIONS="--create-prefs --max-children 5 --username spamd --helper-home-dir /home/spamd/ -s /home/spamd/spamd.log"

# Pid file
# Where should spamd write its PID to file? If you use the -u or
# --username option above, this needs to be writable by that user.
# Otherwise, the init script will not be able to shut spamd down.
PIDFILE="/var/run/spamd.pid"

# Set nice level of spamd
#NICE="--nicelevel 15"

# Cronjob
# Set to anything but 0 to enable the cron job to automatically update
# spamassassin's rules on a nightly basis
CRON=1

```

Figure. Configuration file for `spamassassin`

- Filter settings:

- `mv /etc/spamassassin/local.cf /etc/spamassassin/local.cf.bk`
- `nano /etc/spamassassin/local.cf`

```
rewrite_header Subject ***** SPAM SCORE *****
report_safe 0
required_score 5.0
use_bayes 1
use_bayes_rules 1
bayes_auto_learn 1
skip_rbl_checks 0
use_razor2 0
use_pyzor 0
ifplugin Mail::SpamAssassin::Plugin::Shortcircuit
endif
```

- Restart `systemctl restart postfix`

```
root@mail:/etc/spamassassin# systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/lib/systemd/system/postfix.service; enabled; vendor preset: enabled)
   Active: active (exited) since Tue 2025-12-16 21:02:33 CET; 5s ago
     Docs: man:postfix(1)
   Process: 16285 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
   Main PID: 16285 (code=exited, status=0/SUCCESS)
      CPU: 3ms

Dec 16 21:02:33 mail.nekhaynikita.ru systemd[1]: Starting Postfix Mail Transport Agent...
Dec 16 21:02:33 mail.nekhaynikita.ru systemd[1]: Finished Postfix Mail Transport Agent.
root@mail:/etc/spamassassin# ^C
root@mail:/etc/spamassassin# systemctl status spamassassin
● spamassassin.service - Perl-based spam filter using text analysis
   Loaded: loaded (/lib/systemd/system/spamassassin.service; enabled; vendor preset: enabled)
   Active: active (running) since Tue 2025-12-16 21:02:29 CET; 16s ago
     Process: 15950 ExecStart=/usr/sbin/spamd -d --pidfile=/run/spamd.pid $OPTIONS (code=exited, status=0/SUCCESS)
   Main PID: 15952 (spamd)
      Tasks: 3 (limit: 2154)
     Memory: 182.4M
          CPU: 2.500s
   CGroup: /system.slice/spamassassin.service
           └─15952 /usr/bin/perl "-T -w" /usr/sbin/spamd -d --pidfile=/run/spamd.pid --create-prefs --max-children 5 --username spa
             └─15954 "spamd child"
             └─15955 "spamd child"

Dec 16 21:02:26 mail.nekhaynikita.ru systemd[1]: Stopped Perl-based spam filter using text analysis.
Dec 16 21:02:26 mail.nekhaynikita.ru systemd[1]: spamassassin.service: Consumed 2.954s CPU time.
Dec 16 21:02:26 mail.nekhaynikita.ru systemd[1]: Starting Perl-based spam filter using text analysis...
Dec 16 21:02:26 mail.nekhaynikita.ru systemd[1]: Started Perl-based spam filter using text analysis.
```

Figure. Status of reruned service daemons

- Check via `echo "test" | mail -s "normal message" user1@nekhaynikita.ru`

- It received without a tag
- Check via mail -s "spam test" user1@nekhaynikita.ru < gtube.txt
 - gtube.txt

```
XJS*C4JDBQADN1.NSBN3*2IDNEN*GTUBE-STANDARD-ANTI-UBE-T  
EST-EMAIL*C.34X
```

- In the topic of the message: [SPAM] spamtest
- In logs `tail -f /var/log/mail.log` : spamd: identified spam

- Restarted the services:

```
systemctl start spamassassin.service  
systemctl restart postfix.service
```

- Received error on `spamassassin` :
 - Here I see that SpamAssassin is used via **pipe**, but **spamd is present**, which is not **SMTP/LMTP server**, so Postfix can't communicate it via socket directly.



Dec 16 21:17:29 mail postfix/qmgr[17873]: warning: private/spamassassin socket: malformed response
Dec 16 21:17:29 mail postfix/qmgr[17873]: warning: private/spamassassin socket: malformed response
Dec 16 21:17:29 mail postfix/qmgr[17873]: warning: transport spamassassin failure -- see a previous warning/fatal/panic logfile record for the problem description
Dec 16 21:17:29 mail postfix/qmgr[17873]: warning: transport spamassassin failure -- see a previous warning/fatal/panic logfile record for the problem description
Dec 16 21:17:29 mail postfix/master[17870]: warning: process /usr/lib/postfix/sbin/pipe pid 17889 exit status 1
Dec 16 21:17:29 mail postfix/master[17870]: warning: process /usr/lib/postfix/sbin/pipe pid 17889 exit status 1
Dec 16 21:17:29 mail postfix/master[17870]: warning: /usr/lib/postfix/sbin/pipe: bad command startup -- throttling
Dec 16 21:17:29 mail postfix/master[17870]: warning: /usr/lib/postfix/sbin/pipe: bad command startup -- throttling

- The desired scheme:

SMTP → Postfix
↓
SpamAssassin (spamc → spamd)
↓
Postfix (reinject)
↓
virtual delivery → Dovecot

- Deleted in master.cf line with -o
- Added line in `main.cf`

```
# Spam blocking
content_filter=spamassassin
```

Figure. Spam blocking line in `main.cf`

- I received error in looping errors:

`postfix` → `spamassassin` → `postfix` → `spamassassin` → `postfix` → `spamassassin` ...

- Added line in `master.cf`

```
# SPAMASSASSIN
spamassassin unix -      n      n      -      -      pipe
 flags=Rq user=spamd argv=/usr/bin/spamc -e
 /usr/sbin/sendmail -oi -f ${sender} -o smtp_generic_maps= -o smtp_destination_concurrency_limit=1
 -o content_filter=filtered ${recipient}

# Filtering to discard loops
filtered unix - - n - - smtp
 -o content_filter=
 -o receive_override_options=no_header_body_checks,no_unknown_recipient_checks,no_milters
```

Figure. Updated configuration for `master.cf`

- Changed this conf

```
v7.2 /etc/spamassassin/local.cf
#rewrite_header Subject ***** SPAM SCORE *****
report_safe 0
required_score 5.0
use_bayes 1
use_bayes_rules 1
bayes_auto_learn 0
skip_rbl_checks 0
use_razor2 0
use_pyzor 0
ifplugin Mail::SpamAssassin::Plugin::Shortcircuit
endif
```

Figure. Updated configuration for `local.cf`

- `grep -R "bayes_auto_learn" /etc/spamassassin` - received 0
- Changed `master.cf` again

```
# SPAMASSASSIN
spamassassin unix -      n      n      -      -      pipe
flags=Rq user=spamd argv=/usr/bin/spamc -f
/usr/sbin/sendmail -oi -f ${sender} -o content_filter=${recipient}
```

Figure. Updated configuration for `master.cf`

- Clean queue of messages `postsuper -d ALL`

```
Dec 16 22:00:14 mail postfix/postfix-script[22495]: starting the Postfix mail system
Dec 16 22:00:15 mail postfix/master[22497]: daemon started -- version 3.6.4, configuration /etc/postfix
Dec 16 22:00:15 mail postfix/master[22497]: daemon started -- version 3.6.4, configuration /etc/postfix
Dec 16 22:00:46 mail postfix/smtpd[22506]: warning: dict_nis_init: NIS domain name not set - NIS lookups disabled
Dec 16 22:00:46 mail postfix/smtpd[22506]: warning: dict_nis_init: NIS domain name not set - NIS lookups disabled
Dec 16 22:00:46 mail postfix/smtpd[22506]: connect from mail-ljl-f182.google.com[209.85.208.182]
Dec 16 22:00:46 mail postfix/smtpd[22506]: connect from mail-ljl-f182.google.com[209.85.208.182]
Dec 16 22:00:47 mail postfix/smtpd[22506]: Anonymous TLS connection established from mail-ljl-f182.google.com[209.85.208.182]:
TLShv1.3 with cipher TLS_AES_128_GCM_SHA256 (128/128 bits) key-exchange X25519 server-signature RSA-PSS (2048 bits) server-digest
SHA256
Dec 16 22:00:47 mail postfix/smtpd[22506]: Anonymous TLS connection established from mail-ljl-f182.google.com[209.85.208.182]:
TLShv1.3 with cipher TLS_AES_128_GCM_SHA256 (128/128 bits) key-exchange X25519 server-signature RSA-PSS (2048 bits) server-digest
SHA256
Dec 16 22:00:47 mail postfix/smtpd[22506]: 3365A2087E: client=mail-ljl-f182.google.com[209.85.208.182]
Dec 16 22:00:47 mail postfix/smtpd[22506]: 3365A2087E: client=mail-ljl-f182.google.com[209.85.208.182]
Dec 16 22:00:47 mail postfix/cleanup[22512]: 3365A2087E: message-id=<CAN2vgRyDpRNNF7iCjwXd23T808WHWRULOJEdzSDSf9K8RtRGMA@mail.
mail.com>
Dec 16 22:00:47 mail postfix/cleanup[22512]: 3365A2087E: message-id=<CAN2vgRyDpRNNF7iCjwXd23T808WHWRULOJEdzSDSf9K8RtRGMA@mail.
mail.com>
Dec 16 22:00:47 mail opendkim[12965]: 3365A2087E: mail-ljl-f182.google.com [209.85.208.182] not internal
Dec 16 22:00:47 mail opendkim[12965]: 3365A2087E: not authenticated
Dec 16 22:00:47 mail opendkim[12965]: 3365A2087E: no signing domain match for 'gmail.com'
Dec 16 22:00:47 mail opendkim[12965]: 3365A2087E: no signing subdomain match for 'gmail.com'
Dec 16 22:00:47 mail opendkim[12965]: 3365A2087E: mail-ljl-f182.google.com [209.85.208.182] not internal
Dec 16 22:00:47 mail opendkim[12965]: 3365A2087E: not authenticated
Dec 16 22:00:47 mail opendkim[12965]: 3365A2087E: no signing domain match for 'gmail.com'
Dec 16 22:00:47 mail opendkim[12965]: 3365A2087E: no signing subdomain match for 'gmail.com'
Dec 16 22:00:47 mail opendkim[12965]: 3365A2087E: DKIM verification successful
Dec 16 22:00:47 mail opendkim[12965]: 3365A2087E: DKIM verification successful
Dec 16 22:00:47 mail opendkim[12965]: 3365A2087E: s=20230601 d=gmail.com a=rsa-sha256 SSL
Dec 16 22:00:47 mail opendkim[12965]: 3365A2087E: s=20230601 d=gmail.com a=rsa-sha256 SSL
Dec 16 22:00:47 mail postfix/qmgr[22500]: 3365A2087E: from=<ktofreesapiens@gmail.com>, size=3653, nrcpt=1 (queue active)
Dec 16 22:00:47 mail postfix/qmgr[22500]: 3365A2087E: from=<ktofreesapiens@gmail.com>, size=3653, nrcpt=1 (queue active)
Dec 16 22:00:47 mail postfix/smtpd[22506]: disconnect from mail-ljl-f182.google.com[209.85.208.182] ehlo=2 starttls=1 mail=1 re
pt=1 bdat=1 quit=1 commands=7
Dec 16 22:00:47 mail postfix/smtpd[22506]: disconnect from mail-ljl-f182.google.com[209.85.208.182] ehlo=2 starttls=1 mail=1 re
pt=1 bdat=1 quit=1 commands=7
Dec 16 22:00:47 mail postfix/pipe[22513]: 3365A2087E: to=<user1@nekhaynikita.ru>, relay=spamassassin, delay=0.41, delays=0.18/0
.01/0/0.23, dsn=2.0.0, status=sent (delivered via spamassassin service (Return-Path: <ktofreesapiens@gmail.com> X-Spam-Checker-
Version: SpamAssassin 3.4.6 (2021-04-09) on m))
Dec 16 22:00:47 mail postfix/pipe[22513]: 3365A2087E: to=<user1@nekhaynikita.ru>, relay=spamassassin, delay=0.41, delays=0.18/0
.01/0/0.23, dsn=2.0.0, status=sent (delivered via spamassassin service (Return-Path: <ktofreesapiens@gmail.com> X-Spam-Checker-
Version: SpamAssassin 3.4.6 (2021-04-09) on m))
Dec 16 22:00:47 mail postfix/qmgr[22500]: 3365A2087E: removed
Dec 16 22:00:47 mail postfix/qmgr[22500]: 3365A2087E: removed
```

Figure. Mail log configuration errors resolved

- Check services running and ports listening to spamd and also testing spam message

```

root@mail:~# ss -lntp | grep spamd
LISTEN 0      4096      127.0.0.1:783      0.0.0.0:*      users:((("spamd child",pid=18797,fd=6),("spamd child",pid=18796,fd=6),("spamd",pid=18795,fd=6))
LISTEN 0      4096      :::1:783          ::::*      users:((("spamd child",pid=18797,fd=5),("spamd child",pid=18796,fd=5),("spamd",pid=18795,fd=5))
root@mail:~# echo "XJS*C4JDBQADN1.NSBN3*2IDNEN*GTUBE-STANDARD-ANTI-UBE-TEST-EMAIL*C.34X" | spamc
X-Spam-Checker-Version: SpamAssassin 3.4.6 (2021-04-09) on mail.nekhaynikita.ru
X-Spam-Flag: YES
X-Spam-Level: *****
X-Spam-Status: Yes, score=7.9 required=5.0 tests=EMPTY_MESSAGE,MISSING_DATE,MISSING_FROM,MISSING_HEADERS,MISSING_MID,MISSING_SUBJECT,NO_HEADERS_MESSAGE,NO_RECEIVED,NO_RELAYS autolearn=no autolearn_force=no version=3.4.6
X-Spam-Report:
* -0.0 NO_RELAYS Informational: message was not relayed via SMTP
* 1.2 MISSING_HEADERS Missing To: header
* 2.3 EMPTY_MESSAGE Message appears to have no textual parts and no
* Subject: text
* 0.1 MISSING_MID Missing Message-Id: header
* 1.4 MISSING_DATE Missing Date: header
* -0.0 NO_RECEIVED Informational: message has no Received headers
* 1.8 MISSING_SUBJECT Missing Subject: header
* 1.0 MISSING_FROM Missing From: header
* 0.0 NO_HEADERS_MESSAGE Message appears to be missing most RFC-822
* headers
XJS*C4JDBQADN1.NSBN3*2IDNEN*GTUBE-STANDARD-ANTI-UBE-TEST-EMAIL*C.34X
Subject: ***** SPAM SCORE *****
X-Spam-Prev-Subject: (nonexistent)

Dec 16 21:30:18 mail postfix/pickup[18491]: 7A6C420886: uid=1001 from=<ktofreesapiens@gmail.com>
Dec 16 21:30:18 mail postfix/pickup[18491]: 7A6C420886: uid=1001 from=<ktofreesapiens@gmail.com>
Dec 16 21:30:18 mail postfix/pipe[18834]: 3733E20F36: to=<user1@nekhaynikita.ru>, relay=spamassassin, delay=0.28, delays=0.08/0/0/0.2, dsn=2.0.0, status=sent (delivered via spamassassin service)
Dec 16 21:30:18 mail postfix/pipe[18834]: 3733E20F36: to=<user1@nekhaynikita.ru>, relay=spamassassin, delay=0.28, delays=0.08/0/0/0.2, dsn=2.0.0, status=sent (delivered via spamassassin service)
Dec 16 21:30:18 mail postfix/qmgr[18492]: 3733E20F36: removed
Dec 16 21:30:18 mail postfix/qmgr[18492]: 3733E20F36: removed

```

Figure. Mail logs showing SpamAssassin successfully identifying and tagging spam messages.

Part 8: Troubleshooting methods to solve errors

- Low-level tools were used to:
 - test SMTP and IMAP protocol behavior,
 - validate TLS negotiation,
 - confirm authentication mechanisms.
- Log analysis was essential to:
 - identify configuration loops,
 - diagnose authentication failures,
 - resolve DNS and DKIM mismatches.

Step 8.1: Test SMTP with Telnet

- Tested basic SMTP connectivity:


```
telnet localhost 25
```

- Commands to execute:

```
EHLO localhost
MAIL FROM:user1@example.com
RCPT TO:user2@example.com
DATA
Subject: Test Email
This is a test email.
.
QUIT
```

```

root@mail:~# telnet localhost 25
Trying 127.0.0.1...
Connected to localhost.
Escape character is '^]'.
220 mail.nekhaynikita.ru ESMTP Postfix
EHLO mail.example.com
MAIL FROM:<user1@example.com>
RCPT TO:<user2@example.com>
DATA
Subject: local test

This is a test
.
QUIT250-mail.nekhaynikita.ru
250-PIPELINING
250-SIZE 10240000
250-VERFY
250-ETRN
250-STARTTLS
250-ENHANCEDSTATUSCODES
250-8BITMIME
250-DSN
250 CHUNKING
250 2.1.0 Ok
250 2.1.5 Ok
354 End data with <CR><LF>.<CR><LF>
250 2.0.0 Ok: queued as 1E86220E3C

```

Figure. SMTP commands

- Here I received an error "*505.1.1 user2@example.com: Recipient address rejected: User unknown in local recipient table*" occurs because Postfix is treating "example.com" as a local domain instead of a virtual domain. As a result, it's checking for "user2" as a system user (e.g., in /etc/passwd), which doesn't exist.
 - In my Postfix configuration (/etc/postfix/main.cf) I had:


```
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
```
 - So I changed to:


```
mydestination = $myhostname, localhost.$mydomain, localhost
```

- But this is not a good approach — this is a root problem, so later I fixed it

```
RCPT TO:user2@example.com
501 5.5.4 Syntax: RCPT TO:<address>
RCPT TO:user2@example.com
550 5.1.1 <user2@example.com>: Recipient address rejected: User unknown in local recipient table
```

Figure. Error on SMTP commands

Step 8.2: Test SMTP with TLS

```
openssl s_client -connect localhost:587 -starttls smtp
```

- After connection, test SMTP commands.

Step 8.3: Test IMAP Connection

- Now I test SMTP with TLS

```
openssl s_client -connect localhost:993
```

```
-----END CERTIFICATE-----
subject=CN = mail.nekhaynikita.ru
issuer=C = US, O = Let's Encrypt, CN = R12
---
No client certificate CA names sent
Peer signing digest: SHA256
Peer signature type: RSA-PSS
Server Temp Key: X25519, 253 bits
---
SSL handshake has read 3373 bytes and written 406 bytes
Verification: OK
---
New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384
Server public key is 2048 bit
Secure Renegotiation IS NOT supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
Early data was not sent
Verify return code: 0 (ok)
---
250 CHUNKING
```

Figure. SMTP with TLS log

- I test IMAP connection

```
openssl s_client -connect localhost:993
```

```
a001 LOGIN user1@example.com password123
```

```
a002 LIST "" "*"
```

```
a003 SELECT Sent
```

```
a004 LOGOUT
```

```
* OK [CAPABILITY IMAP4rev1 SASL-IR LOGIN-REFERRALS ID ENABLE IDLE LITER
AL+ AUTH=PLAIN AUTH=LOGIN] Dovecot (Ubuntu) ready.
a001 LOGIN user1@nekhaynikita.ru password123
a001 OK [CAPABILITY IMAP4rev1 SASL-IR LOGIN-REFERRALS ID ENABLE IDLE SO
RT SORT=DISPLAY THREAD=REFERENCES THREAD=REFS THREAD=ORDEREDSUBJECT MUL
TIAPPEND URL-PARTIAL CATENATE UNSELECT CHILDREN NAMESPACE UIDPLUS LIST-
EXTENDED I18NLEVEL=1 CONDSTORE QRESYNC ESEARCH ESORT SEARCHRES WITHIN C
ONTEXT=SEARCH LIST-STATUS BINARY MOVE SNIPPET=FUZZY PREVIEW=FUZZY PREVI
EW STATUS=SIZE SAVEDATE LITERAL+ NOTIFY SPECIAL-USE] Logged in
a002 LIST "" "*"
```

```
* LIST (\HasNoChildren \Sent) "." Sent
* LIST (\HasNoChildren \Trash) "." Trash
* LIST (\HasNoChildren) "." INBOX
a002 OK List completed (0.001 + 0.000 + 0.001 secs).
```

```
a004 SELECT Sent
* OK [CLOSED] Previous mailbox closed.
* FLAGS (\Answered \Flagged \Deleted \Seen \Draft)
* OK [PERMANENTFLAGS (\Answered \Flagged \Deleted \Seen \Draft \*)] Fla
gs permitted.
* 6 EXISTS
* 0 RECENT
* OK [UIDVALIDITY 1765891942] UIDs valid
* OK [UIDNEXT 7] Predicted next UID
a004 OK [READ-WRITE] Select completed (0.001 + 0.000 secs).
```

Figure. IMAP connection log

Step 8.5: Check Mail Logs

- View Postfix logs:

```
sudo tail -f /var/log/mail.log
```

Step 8.6: Verify Email Authentication

- Send an email to an external address (Gmail, etc.) and check headers for:
 - SPF: PASS
 - DKIM: PASS
 - DMARC: PASS

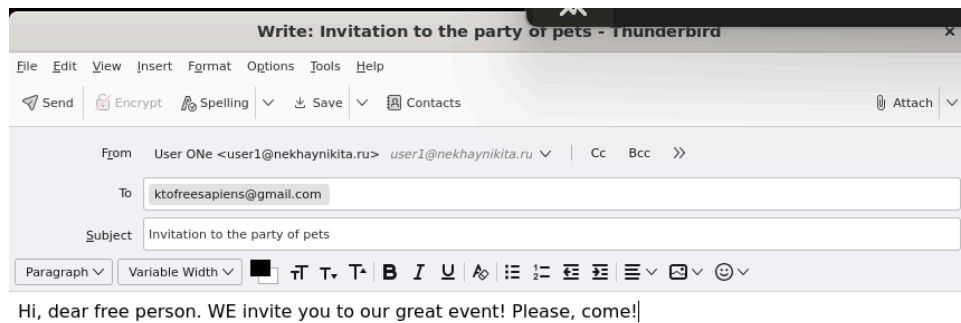


Figure. Email sending from Thunderbird; Sending email to gmail

```
Dec 16 19:20:36 mail postfix/submission/smtpd[10264]: connect from mail.nekhaynikita.ru[91.219.23.38]
Dec 16 19:20:36 mail postfix/submission/smtpd[10264]: connect from mail.nekhaynikita.ru[91.219.23.38]
Dec 16 19:20:37 mail postfix/submission/smtpd[10264]: Anonymous TLS connection established from mail.nekhaynikita.ru[91.219.23.38]: TL
Sv1.3 with cipher TLS_AES_128_GCM_SHA256 (128/128 bits) key-exchange X25519 server-signature RSA-PSS (2048 bits) server-digest SHA256
Dec 16 19:20:37 mail postfix/submission/smtpd[10264]: Anonymous TLS connection established from mail.nekhaynikita.ru[91.219.23.38]: TL
Sv1.3 with cipher TLS_AES_128_GCM_SHA256 (128/128 bits) key-exchange X25519 server-signature RSA-PSS (2048 bits) server-digest SHA256
Dec 16 19:20:37 mail postfix/submission/smtpd[10264]: 5AEDA2087F: cClient=mail.nekhaynikita.ru[91.219.23.38], sasl_method=PLAIN, sasl_u
sername=user1@nekhaynikita.ru
Dec 16 19:20:37 mail postfix/submission/smtpd[10264]: 5AEDA2087F: cClient=mail.nekhaynikita.ru[91.219.23.38], sasl_method=PLAIN, sasl_u
sername=user1@nekhaynikita.ru
Dec 16 19:20:37 mail postfix/cleanup[10323]: 5AEDA2087F: message-id=<ca39c2d4-8a91-474a-a6c7-2630e2d4ce7f@nekhaynikita.ru>
Dec 16 19:20:37 mail postfix/cleanup[10323]: 5AEDA2087F: message-id=<ca39c2d4-8a91-474a-a6c7-2630e2d4ce7f@nekhaynikita.ru>
Dec 16 19:20:37 mail opendkim[1091]: 5AEDA2087F: DKIM-Signature field added (s=default, d=nekhaynikita.ru)
Dec 16 19:20:37 mail opendkim[1091]: 5AEDA2087F: DKIM-Signature field added (s=default, d=nekhaynikita.ru)
Dec 16 19:20:37 mail postfix/qmgr[10085]: 5AEDA2087F: from=<user1@nekhaynikita.ru>, size=883, nrcpt=1 (queue active)
Dec 16 19:20:37 mail postfix/qmgr[10085]: 5AEDA2087F: from=<user1@nekhaynikita.ru>, size=883, nrcpt=1 (queue active)
Dec 16 19:20:37 mail dovecot: imap(user1@nekhaynikita.ru)<10247><u/GwBVGplb2xcm>: Disconnected: Logged out in=625 out=835 deleted=0
expunged=0 trashed=0 hdr_count=0 hdr_bytes=0 body_count=0 body_bytes=0
Dec 16 19:20:37 mail dovecot: imap(user1@nekhaynikita.ru)<10247><u/GwBVGplb2xcm>: Disconnected: Logged out in=625 out=835 deleted=0
expunged=0 trashed=0 hdr_count=0 hdr_bytes=0 body_count=0 body_bytes=0
Dec 16 19:20:37 mail postfix/smtp[10324]: Untrusted TLS connection established to gmail-smtp-in.l.google.com[64.233.184.26]:25: TLSv1.
3 with cipher TLS_AES_256_GCM_SHA384 (256/256 bits) key-exchange X25519 server-signature ECDSA (P-256) server-digest SHA256
Dec 16 19:20:37 mail postfix/smtp[10324]: Untrusted TLS connection established to gmail-smtp-in.l.google.com[64.233.184.26]:25: TLSv1.
3 with cipher TLS_AES_256_GCM_SHA384 (256/256 bits) key-exchange X25519 server-signature ECDSA (P-256) server-digest SHA256
Dec 16 19:20:38 mail postfix/smtp[10324]: 5AEDA2087F: to=<ktofreapiens@gmail.com>, relay=gmail-smtp-in.l.google.com[64.233.184.26]:2
5, delay=1.4, delays=0.15/0.06/0.49/0.67, dsn=2.0.0, status=sent (250 2.0.0 OK 1765909238 ffacd0b85a97d-4318ae0d254si283884f8f.522 -
gsmtpt)
Dec 16 19:20:38 mail postfix/smtp[10324]: 5AEDA2087F: to=<ktofreapiens@gmail.com>, relay=gmail-smtp-in.l.google.com[64.233.184.26]:2
5, delay=1.4, delays=0.15/0.06/0.49/0.67, dsn=2.0.0, status=sent (250 2.0.0 OK 1765909238 ffacd0b85a97d-4318ae0d254si283884f8f.522 -
gsmtpt)
Dec 16 19:20:38 mail postfix/qmgr[10085]: 5AEDA2087F: removed
Dec 16 19:20:38 mail postfix/qmgr[10085]: 5AEDA2087F: removed
Dec 16 19:20:42 mail postfix/submission/smtpd[10264]: disconnect from mail.nekhaynikita.ru[91.219.23.38] ehlo=2 starttls=1 auth=1 mail
=1 rcpt=1 data=1 quit=1 commands=8
Dec 16 19:20:42 mail postfix/submission/smtpd[10264]: disconnect from mail.nekhaynikita.ru[91.219.23.38] ehlo=2 starttls=1 auth=1 mail
=1 rcpt=1 data=1 quit=1 commands=8
```

Figure. Logs of Email sending from Thunderbird



Figure. Email message was received

```

Delivered-To: ktorefreesapiens@gmail.com
Received: by 2002:aa6:cd0a:0:b0:325:a74d:226 with SMTP id i10csp27459861kb;
    Tue, 16 Dec 2025 10:20:38 -0800 (PST)
X-Google-Smtp-Source: AGHT+IEewlh/zE4yfpq+qG5FY0xyl2/Ya20o8vzmu1SWMdpOSsDj4tcxFKkgmRzajZ3WmbKCo5IH
X-Received: by 2002:a05:6000:438a:b0:430:fa52:9ea9 with SMTP id
    ffacd0b85a97d-430fa52a0f6mr11121505f8f.62.1765909238516;
    Tue, 16 Dec 2025 10:20:38 -0800 (PST)
ARC-Seal: i=1; a=rsa-sha256; t=1765909238; cv=none;
    b=google.com; s=arc-20240605;
    b=OQ6BdcsMlwrtmNDYwd+tGN5XDvR+A+v1/vERHYSTU0mUD1fm6ZfoCNW0FZL5Do4FRjg
    +xEgZm4uYrHHeibT/Mq19MdzePX8P8pUx9CXkYOHAV+mGPDudRock1tnwZysFJe+F2e3
    eTm+Z48Nh4NkF5omojrP0Cc6W4W3hbNJSQfLkpg3gErOfGtoxffkDyJ7YuGvKg4M0aas
    O3vMCooHCLK/yakG8axCFI6RIWsCgLSkhohbEKV64atxwZKrI+qCa3em+Em5u+skU8C
    GgL9Kmllee7VooyTa1KvXVSs1uSwGePzA6RbL4c5wf+F+FmaVbP4582XFTIX9DOKLaw0r
    V02w==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20240605;
    h=content-transfer-encoding:subject:from:to:content-language
    :user-agent:mime-version:date:message-id:dkim-signature;
    bh=zYgXTPcEQhCj4PegZGSqi3WkVQ6/I1q10Dga98vLaY=;
    fh=bfaHt+TCpSGGFwrvP+F1xE0DGOfkmiIYFR6Jrbm60=;
    b=WrWt8nQdH66zEyG0p5/yJLHjYVv4CL8Ik0GINxe9o8rE3tnAPUQ0/Q0y70CjzEugj1
    2m6LGMNBy327cmypzpjht7fO5QddjmfT6g3bU0yNNDhgjC4RhJCw1FwWzjEYr7geg0f
    6h5xepofdl1lONuFa1HPOsy4tg6gJ5FW5ZpV6/GCZqRgs7MfcaBxeXlfnyvQHHSvJr
    DIVFdPhLp5K/clWgYz0AkyEt2iQobCO98wHy9MKGP6uWsA3pIh1MyOsYonRsH2tLmBfk
    INgdyd/iY1oLEcV/3OI77at7etOpsWfGyQINkPM8/B7kqsz9+Twq2+eVOW97EonNu5
    nk/A==;
    dara=google.com
ARC-Authentication-Results: i=1; mx.google.com;
    dkim=neutral (invalid public key) header.i=@nekhaynikita.ru header.s=default header.b=CNDYJjUu;
    spf=pass (google.com: domain of user1@nekhaynikita.ru designates 91.219.23.38 as permitted sender)
    smtp.mailfrom=user1@nekhaynikita.ru;
    ----- (p=QUARANTINE sp=QUARANTINE dis=NONE) header.from=nekhaynikita.ru

dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) header.from=nekhaynikita.ru
Return-Path: <user1@nekhaynikita.ru>
Received: from mail.nekhaynikita.ru (mail.nekhaynikita.ru. [91.219.23.38])
    by mx.google.com with ESMTPS id ffacd0b85a97d-4310ae0d254si283884f8f.522.2025.12.16.10.20.38
    for <ktorefreesapiens@gmail.com>
    (version=TLSv1.3 cipher=TLS_AES_256_GCM_SHA384 bits=256/256);
    Tue, 16 Dec 2025 10:20:38 -0800 (PST)
Received-SPF: pass (google.com: domain of user1@nekhaynikita.ru designates 91.219.23.38 as permitted sender) client-ip=91.219.23.38;
Authentication-Results: mx.google.com;
    dkim=neutral (invalid public key) header.i=@nekhaynikita.ru header.s=default header.b=CNDYJjUu;
    spf=pass (google.com: domain of user1@nekhaynikita.ru designates 91.219.23.38 as permitted sender)
    smtp.mailfrom=user1@nekhaynikita.ru;
    dmarc=pass (p=QUARANTINE sp=QUARANTINE dis=NONE) header.from=nekhaynikita.ru
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/simple; d=nekhaynikita.ru; s=default; t=1765909237; bh=zYgXTPcEQhCj4PegZGSqi3WkVQ6/
    I1q10Dga98vLaY=; h=Date:To:From:Subject:From; b=CNDYJjUu4D7gtLhL/NF9eDCsNA2D2ZuQ0PX7Mx5getHBmyxNCaWeODS570dw4154
    7eGXmWk9+M4Z1LAooe/HcrOgk893PM5c8Dj2pWlRnI/p/35yIpXNPEHuQ3KKnzDcz/
    thRXmEdyAlxa2acMXEg0ME19p6Ek6ymQYU1uuBiosQIE8E5KeAPRwpp+MX4ura0e9
    70BciHsq4NKa2e54C6DuW15xEegh10+ss9bbdMi6/N4iyQtCe7RmcQ+pPyCz44wQL
    P21H1IWG4Ap0gomkyGPALNQ4DdRmEhu0iZxLPhkQkdjziJmERsvfBxm+Qdkz7mXgc0
    8kU1SutVEDBuQ==
Received: from [91.219.23.38] (mail.nekhaynikita.ru [91.219.23.38]) (using TLSv1.3 with cipher TLS_AES_128_GCM_SHA256 (128/128 bits)
    key-exchange X25519 server-signature RSA-PSS (2048 bits) server-digest SHA256) (No client certificate requested) by
    mail.nekhaynikita.ru (Postfix) with ESMTPSA id 5AEDA2087F for <ktorefreesapiens@gmail.com>; Tue, 16 Dec 2025 19:20:37 +0100 (CET)
Message-ID: <ca39c2d4-8a91-474a-a6c7-2630e2d4ce7f@nekhaynikita.ru>
Date: Tue, 16 Dec 2025 19:20:36 +0100
MIME-Version: 1.0
User-Agent: Mozilla Thunderbird
Content-Language: en-US
To: ktorefreesapiens@gmail.com
From: User One <user1@nekhaynikita.ru>
Subject: Invitation to the party of pets
Content-Type: text/plain; charset=UTF-8; format=flowed
Content-Transfer-Encoding: 7bit

Hi, dear free person. WE invite you to our great event! Please, come!

```

Figure. Headers and security of message

```

Dec 16 19:23:20 mail postfix/smtpd[10354]: warning: dict_nis_init: NIS domain name not set - NIS lookups disabled
Dec 16 19:23:20 mail postfix/smtpd[10354]: warning: dict_nis_init: NIS domain name not set - NIS lookups disabled
Dec 16 19:23:20 mail postfix/smtpd[10354]: connect from mail-lf1-f48.google.com[209.85.167.48]
Dec 16 19:23:20 mail postfix/smtpd[10354]: Anonymous TLS connection established from mail-lf1-f48.google.com[209.85.167.48]: TLSv1.3 w
ith cipher TLS_AES_128_GCM_SHA256 (128/128 bits) key-exchange X25519 server-signature RSA-PSS (2048 bits) server-digest SHA256
Dec 16 19:23:20 mail postfix/smtpd[10354]: Anonymous TLS connection established from mail-lf1-f48.google.com[209.85.167.48]: TLSv1.3 w
ith cipher TLS_AES_128_GCM_SHA256 (128/128 bits) key-exchange X25519 server-signature RSA-PSS (2048 bits) server-digest SHA256
Dec 16 19:23:20 mail postfix/smtpd[10354]: E436E2087F: client=mail-lf1-f48.google.com[209.85.167.48]
Dec 16 19:23:20 mail postfix/smtpd[10354]: E436E2087F: client=mail-lf1-f48.google.com[209.85.167.48]
Dec 16 19:23:20 mail postfix/cleanup[10358]: E436E2087F: message-id=<CAN2vgRwjhQC3iRknMcr8n=LjV0msLPt=NgdFJDrg_1Dhcm8=uw@mail.gmail.co
m>
Dec 16 19:23:20 mail postfix/cleanup[10358]: E436E2087F: message-id=<CAN2vgRwjhQC3iRknMcr8n=LjV0msLPt=NgdFJDrg_1Dhcm8=uw@mail.gmail.co
m>
Dec 16 19:23:20 mail opendkim[1091]: E436E2087F: mail-lf1-f48.google.com [209.85.167.48] not internal
Dec 16 19:23:20 mail opendkim[1091]: E436E2087F: mail-lf1-f48.google.com [209.85.167.48] not internal
Dec 16 19:23:20 mail opendkim[1091]: E436E2087F: not authenticated
Dec 16 19:23:20 mail opendkim[1091]: E436E2087F: not authenticated
Dec 16 19:23:20 mail opendkim[1091]: E436E2087F: no signing domain match for 'gmail.com'
Dec 16 19:23:20 mail opendkim[1091]: E436E2087F: no signing domain match for 'gmail.com'
Dec 16 19:23:20 mail opendkim[1091]: E436E2087F: no signing subdomain match for 'gmail.com'
Dec 16 19:23:20 mail opendkim[1091]: E436E2087F: no signing subdomain match for 'gmail.com'
Dec 16 19:23:21 mail opendkim[1091]: E436E2087F: DKIM verification successful
Dec 16 19:23:21 mail opendkim[1091]: E436E2087F: DKIM verification successful
Dec 16 19:23:21 mail opendkim[1091]: E436E2087F: s=20230601 d=gmail.com a=rsa-sha256 SSL
Dec 16 19:23:21 mail opendkim[1091]: E436E2087F: s=20230601 d=gmail.com a=rsa-sha256 SSL
Dec 16 19:23:21 mail postfix/qmgr[10085]: E436E2087F: from=<ktofreesapiens@gmail.com>, size=4561, nrcpt=1 (queue active)
Dec 16 19:23:21 mail postfix/qmgr[10085]: E436E2087F: from=<ktofreesapiens@gmail.com>, size=4561, nrcpt=1 (queue active)
Dec 16 19:23:21 mail postfix/virtual[10360]: E436E2087F: to=<user1@nekhaynikita.ru>, relay=virtual, delay=0.25, delays=0.23/0.01/0/0,
dsn=2.0.0, status=sent (delivered to maildir)
Dec 16 19:23:21 mail postfix/virtual[10360]: E436E2087F: to=<user1@nekhaynikita.ru>, relay=virtual, delay=0.25, delays=0.23/0.01/0/0,
dsn=2.0.0, status=sent (delivered to maildir)
Dec 16 19:23:21 mail postfix/qmgr[10085]: E436E2087F: removed
Dec 16 19:23:21 mail postfix/qmgr[10085]: E436E2087F: removed
Dec 16 19:23:21 mail postfix/smtpd[10354]: disconnect from mail-lf1-f48.google.com[209.85.167.48] ehlo=2 starttls=1 mail=1 rcpt=1 bdat
=1 quit=1 commands=7
Dec 16 19:23:21 mail postfix/smtpd[10354]: disconnect from mail-lf1-f48.google.com[209.85.167.48] ehlo=2 starttls=1 mail=1 rcpt=1 bdat
=1 quit=1 commands=7

```

Figure. Logs for send message from gmail to Thunderbird

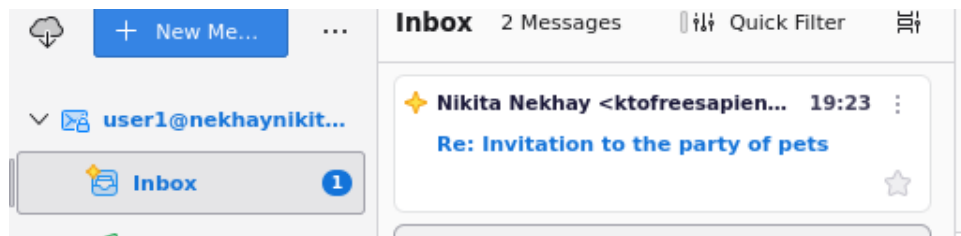


Figure. Receiving email from gmail 1

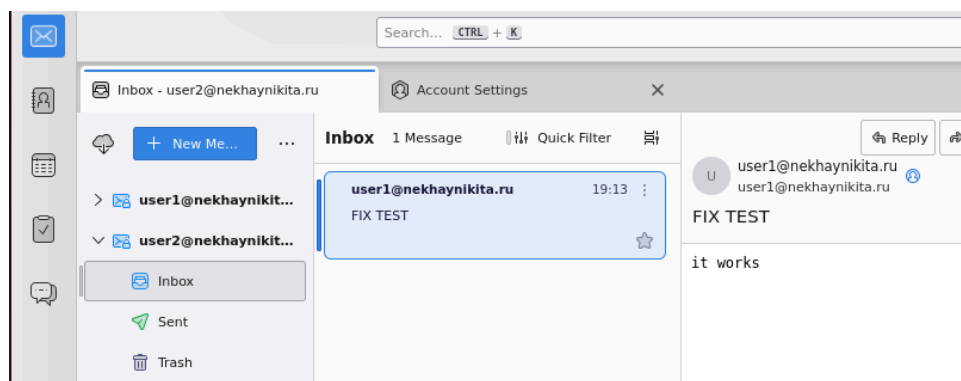


Figure. Receiving email from gmail 1

```

root@mail:~# ls /var/mail/vhosts/nekhaynikita.ru/user2/new
1765908806.Vfc02I683d5M227666.mail.nekhaynikita.ru
root@mail:~# tail -f /var/log/mail.log
Dec 16 19:13:26 mail opendkim[1091]: 0D33F2087D: DKIM-Signature field added (s=default, d=nekhaynikita.ru)
Dec 16 19:13:26 mail opendkim[1091]: 0D33F2087D: DKIM-Signature field added (s=default, d=nekhaynikita.ru)
Dec 16 19:13:26 mail postfix/qmgr[10085]: 0D33F2087D: from=<user1@nekhaynikita.ru>, size=392, nrcpt=1 (queue active)
Dec 16 19:13:26 mail postfix/qmgr[10085]: 0D33F2087D: from=<user1@nekhaynikita.ru>, size=392, nrcpt=1 (queue active)
Dec 16 19:13:26 mail postfix/virtual[10137]: 0D33F2087D: to=<user2@nekhaynikita.ru>, relay=virtual, delay=14, delays=14/0.02/0/0, dsn=2.0.0, status=sent (delivered to maildir)
Dec 16 19:13:26 mail postfix/virtual[10137]: 0D33F2087D: to=<user2@nekhaynikita.ru>, relay=virtual, delay=14, delays=14/0.02/0/0, dsn=2.0.0, status=sent (delivered to maildir)
Dec 16 19:13:26 mail postfix/qmgr[10085]: 0D33F2087D: removed
Dec 16 19:13:26 mail postfix/qmgr[10085]: 0D33F2087D: removed
Dec 16 19:13:28 mail postfix/smtpd[10129]: disconnect from localhost[127.0.0.1] ehlo=1 mail=1 rcpt=1 data=1 quit=1 commands=5
Dec 16 19:13:28 mail postfix/smtpd[10129]: disconnect from localhost[127.0.0.1] ehlo=1 mail=1 rcpt=1 data=1 quit=1 commands=5

```

Figure. Logs of authentication, DKIM: Receiving email from gmail

Source of: imap://user2%40nekhaynikita.ru@mail.nekhaynikita.ru:993/f... x

File Edit View Help

X-Mozilla-Status: 0001
X-Mozilla-Status2: 00000000
Return-Path: <user1@nekhaynikita.ru>
X-Original-To: user2@nekhaynikita.ru
Delivered-To: user2@nekhaynikita.ru
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/simple; d=nekhaynikita.ru; s=default; t=1765908806; bh=I5CiMuZZ1F0tJtj2Z29AGFKwLHbns9S6UvrfACElBmo=; h=Subject:Date:From:From; b=WXU1b8Bg4u9D34Q7yLoWForrgHDh6SuqnZjH7fP6G+oqecYYRkevmZL0m2zcr+aZVhjI4d6RXpLLVDFfwEitYeSkUMW20h/yJcpr+ku35uiWs/WER38Fxe/zYV0k+fWMSHLuiv+uC02VzDyXVGj9ZPBy2bq9Q38oCPFS+kJAKMgkMH18GKCGN2/6UvSRwDhUHwie2fVPaLT6wcxJbmd/A9KE8qP9660KoU0hxVl1l6q9rZkNENrYGzPWyA8BmTSLU6MVG6Hlcq9YuC4eVYGWg/pSViKTPDBcfy5IcuvS5Vm5yW9dnYDa0yNZ4bancHsry51jj0Y7y7DWD3lbRl/29Q==

Received: from mail.nekhaynikita.ru (localhost [127.0.0.1])
by mail.nekhaynikita.ru (Postfix) with ESMTP id 0D33F2087D
for <user2@nekhaynikita.ru>; Tue, 16 Dec 2025 19:13:12 +0100 (CET)

Subject: FIX TEST
Message-Id: <20251216181315.0D33F2087D@mail.nekhaynikita.ru>
Date: Tue, 16 Dec 2025 19:13:12 +0100 (CET)
From: user1@nekhaynikita.ru

it works

Figure. Headers for successful email sending from user1→user2

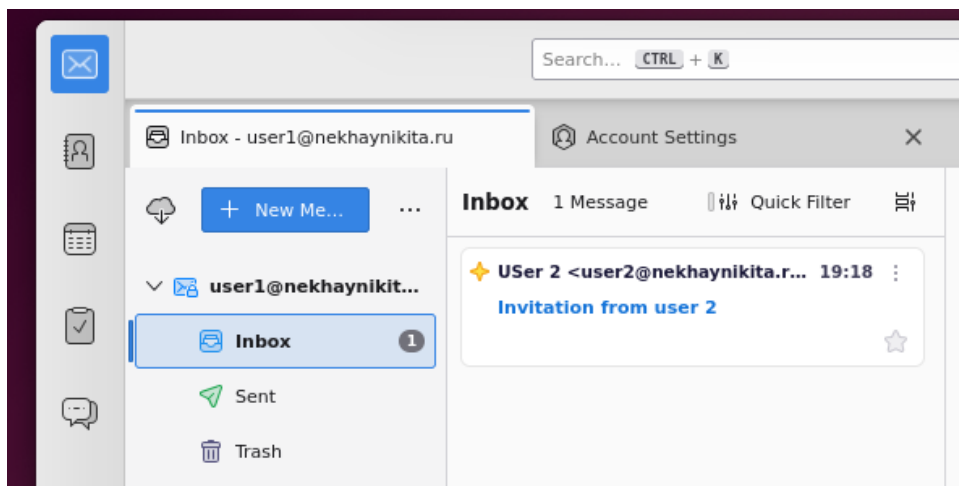


Figure. Email sent from user2→user1 received

```
Dec 16 19:18:52 mail postfix/submission/smtpd[10264]: connect from mail.nekhaynikita.ru[91.219.23.38]
Dec 16 19:18:53 mail postfix/submission/smtpd[10264]: Anonymous TLS connection established from mail.nekhaynikita.ru[91.219.23.38]: TL
Sv1.3 with cipher TLS_AES_128_GCM_SHA256 (128/128 bits) key-exchange X25519 server-signature RSA-PSS (2048 bits) server-digest SHA256
Dec 16 19:18:53 mail postfix/submission/smtpd[10264]: Anonymous TLS connection established from mail.nekhaynikita.ru[91.219.23.38]: TL
Sv1.3 with cipher TLS_AES_128_GCM_SHA256 (128/128 bits) key-exchange X25519 server-signature RSA-PSS (2048 bits) server-digest SHA256
Dec 16 19:18:53 mail postfix/submission/smtpd[10264]: 3E3D920E55: client=mail.nekhaynikita.ru[91.219.23.38], sasl_method=PLAIN, sasl_u
sername=user2@nekhaynikita.ru
Dec 16 19:18:53 mail postfix/submission/smtpd[10264]: 3E3D920E55: client=mail.nekhaynikita.ru[91.219.23.38], sasl_method=PLAIN, sasl_u
sername=user2@nekhaynikita.ru
Dec 16 19:18:53 mail postfix/cleanup[10272]: 3E3D920E55: message-id=<471c7f82-9781-4ad9-b559-91ef7ae75369@nekhaynikita.ru>
Dec 16 19:18:53 mail postfix/cleanup[10272]: 3E3D920E55: message-id=<471c7f82-9781-4ad9-b559-91ef7ae75369@nekhaynikita.ru>
Dec 16 19:18:53 mail opendkim[1091]: 3E3D920E55: DKIM-Signature field added (s=default, d=nekhaynikita.ru)
Dec 16 19:18:53 mail opendkim[1091]: 3E3D920E55: DKIM-Signature field added (s=default, d=nekhaynikita.ru)
Dec 16 19:18:53 mail postfix/qmgr[10085]: 3E3D920E55: from=<user2@nekhaynikita.ru>, size=811, nrcpt=1 (queue active)
Dec 16 19:18:53 mail postfix/qmgr[10085]: 3E3D920E55: from=<user2@nekhaynikita.ru>, size=811, nrcpt=1 (queue active)
Dec 16 19:18:53 mail postfix/virtual[10273]: 3E3D920E55: to=<user1@nekhaynikita.ru>, relay=virtual, delay=0.35, delays=0.33/0.02/0/0,
dsn=2.0.0, status=sent (delivered to maildir)
Dec 16 19:18:53 mail postfix/virtual[10273]: 3E3D920E55: to=<user1@nekhaynikita.ru>, relay=virtual, delay=0.35, delays=0.33/0.02/0/0,
dsn=2.0.0, status=sent (delivered to maildir)
Dec 16 19:18:53 mail postfix/qmgr[10085]: 3E3D920E55: removed
Dec 16 19:18:53 mail postfix/qmgr[10085]: 3E3D920E55: removed
Dec 16 19:18:53 mail dovecot: imap-login: Login: user=<user2@nekhaynikita.ru>, method=PLAIN, rip=91.219.23.38, lip=91.219.23.38, mpid=
10276, TLS, sessions=<d2ewhVGtpb2xcm>
Dec 16 19:18:53 mail dovecot: imap-login: Login: user=<user2@nekhaynikita.ru>, method=PLAIN, rip=91.219.23.38, lip=91.219.23.38, mpid=
10276, TLS, sessions=<d2ewhVGtpb2xcm>
Dec 16 19:18:54 mail dovecot: imap(user2@nekhaynikita.ru)<10276><d2ewhVGtpb2xcm>: Disconnected: Logged out in=486 out=628 deleted=0
expunged=0 trashed=0 hdr_count=0 hdr_bytes=0 body_count=0 body_bytes=0
Dec 16 19:18:54 mail dovecot: imap(user2@nekhaynikita.ru)<10276><d2ewhVGtpb2xcm>: Disconnected: Logged out in=486 out=628 deleted=0
expunged=0 trashed=0 hdr_count=0 hdr_bytes=0 body_count=0 body_bytes=0
Dec 16 19:18:58 mail postfix/submission/smtpd[10264]: disconnect from mail.nekhaynikita.ru[91.219.23.38] ehlo=2 starttls=1 auth=1 mail
```

Figure. Logs for successful email sending from user2→user1

Step 8.7: Test Mail Queue

- Checked mail queue:

```
root@mail:~# mailq
-Queue ID- --Size-- ----Arrival Time---- -Sender/Recipient-----
51AE1422FD      382 Wed Dec 10 21:43:59 root@mail.example.com
(connect to alt4.gmail-smtp-in.l.google.com[108.177.125.27]:25: Connection refused)
ktoofreesapiens@gmail.com

-- 0 Kbytes in 1 Request.
```

Figure. mail queue: logs

Step: 9.1 Check DNS Records

If any fail → my domain/VPS provider is misconfigured.

- A record

dig A mail.nekhaynikita.ru +short

- Saw: **91.219.23.38**

- Check MX record

```
dig MX nekhaynikita.ru +short
```

- Expected: `10 mail.nekhaynikita.ru.`
- But I saw: `10 mx1.timeweb.ru.` — from previous configs.
- So I created it again but for the right domain.

The screenshot shows a DNS management interface with a dark theme. At the top, there's a dropdown menu for 'Домен' (Domain) set to 'Все' (All). To the right are buttons for 'Импортировать записи' (Import records) and 'Добавить запись' (Add record). Below this is a tabbed interface with tabs for 'Все' (All), 'A', 'AAAA', 'TXT', 'SRV', 'CNAME', and 'MX'. The 'MX' tab is selected. A table below shows the MX records:

Тип	Хост	Значение	TTL	Действия
☐ MX	nekhaynikita.ru	10 mail.nekhaynikita.ru	600	...

Figure. MX record inside DNS

- Check SPF

```
dig TXT nekhaynikita.ru +short
```

- Expected to contain: `v=spf1 ip4:203.0.113.10 -all`
- But I saw: different so I changed.

The screenshot shows the same DNS management interface as before, but with the 'TXT' tab selected. The table below shows the TXT records:

Тип	Хост	Значение	TTL	Действия
☐ TXT	nekhaynikita.ru	v=spf1 mx a ip4:91.219.23.38 -all	600	...

Figure. SPF record inside DNS

- Check DKIM

```
dig TXT default._domainkey.nekhaynikita.ru +short
```

- Showed:

```
v=DKIM1; k=rsa; p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8A...
```

- Check DMARC

```
dig TXT _dmarc.nekhaynikita.ru +short
```

- Showed: `"v=DMARC1; p=quarantine; rua=mailto:postmaster@nekhaynikita.ru"`

- Check PTR (reverse DNS)

```
dig -x 91.219.23.38 +short
```

- Expected: mail.nekhaynikita.ru.

- I found root problem:

```
Dec 16 18:07:05 mail postfix/trivial-rewrite[8936]: warning: do not list domain nekhaynikita.ru in BOTH mydestination and virtual_mail_box_domains
Dec 16 18:07:05 mail postfix/trivial-rewrite[8936]: warning: do not list domain nekhaynikita.ru in BOTH mydestination and virtual_mail_box_domains
```

Figure. Log of root problem

- I replaced **ENTIRE mydestination** line with:

```
V7.2 /etc/postfix/main.cf
# Basic Settings
myhostname = mail.nekhaynikita.ru
mydomain = nekhaynikita.ru
myorigin = $mydomain
mydestination = localhost
```

Figure. Solution to root problem.

- I received error about DKIM, so I ensured my DKIM record in domain provider - deleted it and pasted the whole content of `cat /etc/openssl/keys/nekhaynikita.ru/default.txt` to it.
 - I recreated the keys from the zero because physical file and DNS record do not match

```

root@mail:~# opendkim-testkey -d nekhaynikita.ru -s default -vvv
opendkim-testkey: using default configfile /etc/opendkim.conf
opendkim-testkey: key loaded from /etc/opendkim/keys/nekhaynikita.ru/default.private
opendkim-testkey: checking key 'default._domainkey.nekhaynikita.ru'
opendkim-testkey: key not secure
opendkim-testkey: keys do not match

```

Figure. Comparison of DKIM keys is not successful

Error:



Dec 16 18:07:05 mail postfix/smtp[8938]: 8DD3A20E55:
to=ktofreapiens@gmail.com, relay=gmail-smtp-in.l.google.com[64.233.184.26]:25, delay=0.98,
delays=0.18/0.08/0.24/0.48, dsn=5.7.26, status=bounced (host
gmail-smtp-in.l.google.com[64.233.184.26] said: 550-5.7.26 Your
email has been blocked because the sender is unauthenticated. 550-
5.7.26 Gmail requires all senders to authenticate with either SPF or
DKIM. 550-5.7.26 550-5.7.26 Authentication results: 550-5.7.26
DKIM = did not pass 550-5.7.26 SPF [nekhaynikita.ru] with ip:
[91.219.23.38] = did not pass 550-5.7.26 550-5.7.26 For instructions
on setting up authentication, go to 550 5.7.26
<https://support.google.com/mail/answer/81126#authentication>
5b1f17b1804b1-47a8f7179d8si50825885e9.3 - gsmtpt (in reply to end
of DATA command))
Dec 16 18:07:05 mail postfix/smtp[8938]: 8DD3A20E55:
to=ktofreapiens@gmail.com, relay=gmail-smtp-in.l.google.com[64.233.184.26]:25, delay=0.98,
delays=0.18/0.08/0.24/0.48, dsn=5.7.26, status=bounced (host
gmail-smtp-in.l.google.com[64.233.184.26] said: 550-5.7.26 Your
email has been blocked because the sender is unauthenticated. 550-
5.7.26 Gmail requires all senders to authenticate with either SPF or
DKIM. 550-5.7.26 550-5.7.26 Authentication results: 550-5.7.26
DKIM = did not pass 550-5.7.26 SPF [nekhaynikita.ru] with ip:
[91.219.23.38] = did not pass 550-5.7.26 550-5.7.26 For instructions
on setting up authentication, go to 550 5.7.26 <https://suppor>

- But the real cause was me not understating value of `default.txt`: the key is to big, that it divides value into separate `"" "" ""` — but this is the one value of a key.
- Error is fixed:

```
root@mail:~# opendkim-testkey -d nekhaynikita.ru -s default -vvv
opendkim-testkey: using default configfile /etc/opendkim.conf
opendkim-testkey: key loaded from /etc/opendkim/keys/nekhaynikita.ru/default.private
opendkim-testkey: checking key 'default._domainkey.nekhaynikita.ru'
opendkim-testkey: key not secure
opendkim-testkey: key OK
```

Figure. Comparison of DKIM keys is successful

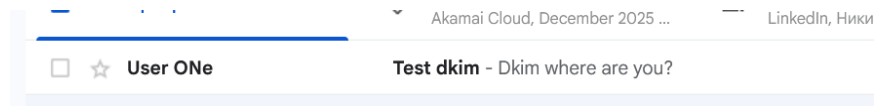


Figure. Testing received messages on DKIM 1

Исходное сообщение	
Идентификатор сообщения	<3228f38b-ae01-48be-9017-9d4437cb204f@nekhaynikita.ru>
Создано:	16 декабря 2025 г. в 22:10 (доставлено через 1 секунду)
От:	User ONE <user1@nekhaynikita.ru>
Кому:	ktofreesapiens@gmail.com
Тема:	Test dkim
SPF:	PASS с IP-адресом 91.219.23.38. Подробнее...
DKIM:	'PASS', домен nekhaynikita.ru Подробнее...
DMARC:	'PASS' Подробнее...

Figure. Testing received messages on DKIM 2

- I received error in logs that `vmailbox.db` is missing, so I created it.

```
root@mail:~# sudo tail -f /var/log/mail.err
Dec 16 14:59:24 mail postfix/submission/smtpd[6013]:
error: open database /etc/postfix/vmailbox.db: No such file
or directory
Dec 16 15:02:18 mail postfix/smtpd[6068]: error: open
database /etc/postfix/vmailbox.db: No such file or
directory
```

Figure. Error for `vmailbox.db`

- Verifying presence of created db.

```
postmap /etc/postfix/vmbox
```

```
ls -l /etc/postfix/vmailbox*
```

- I see: `vmailbox`, `vmailbox.db`

```
root@mail:~# postmap /etc/postfix/vmailbox
root@mail:~# ls -l /etc/postfix/vmailbox*
-rw-r--r-- 1 root root  90 Dec 16 15:14 /etc/postfix/vmailbox
-rw-r--r-- 1 root root 12288 Dec 16 15:15 /etc/postfix/vmailbox.db
root@mail:~#
```

Figure. `ls` logs

Summary

This project demonstrated the end-to-end deployment of a secure, production-style email server using open-source technologies on a Linux VPS. The system integrated Postfix for SMTP mail transport, Dovecot for IMAP access, virtual mailboxes for domain users, and TLS encryption to protect data in transit. Email authentication mechanisms (SPF, DKIM, and DMARC) were implemented and validated to meet modern email security requirements and ensure compatibility with major email providers.

A significant part of the work involved troubleshooting real-world configuration issues, including DNS misalignment, virtual domain handling errors, DKIM key mismatches, and SpamAssassin integration loops. These issues highlighted the tight coupling between mail server configuration, DNS records, and authentication

policies. Resolving them required systematic log analysis, protocol-level testing using OpenSSL and Telnet, and iterative configuration refinement.

The final system successfully delivered encrypted, authenticated, and spam-filtered email between local users and external providers such as Gmail. This confirms that a secure and reliable email infrastructure can be built using Postfix and Dovecot when industry best practices are followed. The project provided practical experience with mail protocols, security mechanisms, and operational troubleshooting that closely reflects real-world system administration tasks.

References

1. Spamassassin installation <https://arenda-server.cloud/blog/kak-nastroit-pochtovyj-server-postfix-s-dovecot/>
2. Postfix Official Documentation.
<https://www.postfix.org/documentation.html>
3. Dovecot IMAP Server Documentation.
<https://doc.dovecot.org/>
4. Let's Encrypt Documentation – TLS Certificates.
<https://letsencrypt.org/docs/>
5. OpenDKIM Project Documentation.
<https://opendkim.org/documentation/>
6. SpamAssassin Official Documentation.
<https://spamassassin.apache.org/documentation/>
7. RFC 5321 – Simple Mail Transfer Protocol (SMTP).
<https://datatracker.ietf.org/doc/html/rfc5321>
8. RFC 6376 – DomainKeys Identified Mail (DKIM).
<https://datatracker.ietf.org/doc/html/rfc6376>
9. RFC 7208 – Sender Policy Framework (SPF).
<https://datatracker.ietf.org/doc/html/rfc7208>

10. RFC 7489 – Domain-based Message Authentication, Reporting, and Conformance (DMARC).

<https://datatracker.ietf.org/doc/html/rfc7489>

11. Mozilla Thunderbird Documentation.

<https://support.mozilla.org/en-US/products/thunderbird>